A_GMX: adaptive Gaussian-mixture clustering for skipper-CCD images

what every step does, the mathematics behind it, and how it performs against WADERs

comm_scripts clustering study

23–25 October 2026

Code: `clustering_tests/adaptive.py` (`adaptive_cluster(..., engine="gmx")`) and `clustering_tests/gmx.py` (`general_cluster`). Figures are produced by `make_agmx_figures.py` (every algorithm step, run on a real image) and `make_agmx_perf.py` (performance). All numbers are recomputed from the saved results in `~/Documents/comm_scripts_data/session_2026-10-23_clustering/clustering_2026-10-25/`.

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1 Summary

What it is. A_GMX clusters a calibrated skipper-CCD image (pixel values in electrons) without deciding pixel by pixel "in or out". It describes the image as **empty background + a sum of Gaussian charge clouds**, fits all clouds at once, shares every pixel's charge among the clouds in proportion to what each predicts there, and then groups clouds into physical deposits with two tests: a geometric one (is there a dip between two clouds?) and a statistical one (is this region several round X-rays, or one irregular deposit?). A single switch in front of it hands images whose background is *not* dark current + read noise to WADERs-like hysteresis clustering.

Why. WADERs' fixed thresholds (10 and 30 times the pedestal sigma) cut the tails of diffuse clusters (energies up to 8 % low on average, 26 % on the worst CCD), merge X-rays closer than ∼6 px, cannot see few-electron events, and on a crowded CCD every threshold is a compromise between merging and cutting.

How it does (details in section 14):

	WADERs	A_GMX
Fe-55 energy of injected clusters, width 0.3 → 1.8 px (median of 8 CCDs)	0.997 → 0.921	1.000 → 1.000
worst CCD at width 1.8 px	0.737	0.994
real Fe-55 Kα (front side): mean / spread over 8 CCDs	0.9928 / 1.32 %	0.9969 / 0.23 %
two diffuse Fe-55 clusters 5 px apart resolved	0.00	0.93
wide test, 878 image-CCDs: Fe-55 / muon / worm / 4-px pair success	0.86 / 0.59 / 0.74 / 0.11	0.93 / 0.64 / 0.79 / 0.88
Fe-55 energy resolution (rms of E/E_{true}), wide test	0.52 %	0.19 %
crowded Am-241 CCD (29 images): Np Lβ / Lγ / γ 26.3 / Ag K significance	15.5 / 16.0 / 14.8 / 12.9 σ	20.8 / 19.2 / 19.9 / 18.1 σ
crowded CCD energy scale k (1 = right)	0.984	1.002
few-electron events, 0.001 e^-/pix: 3-e^- found / fakes per Mpix	0.00 / 0	0.81 / 0
few-electron events, 0.01 e^-/pix: 5-e^- found / fakes per Mpix	0.11 / 0	0.93 / 0
run time per image-CCD (median)	0.01 s	0.7 s

Recommendation (section 21). A_GMX supersedes MHC, the hybrids and the adaptive-MHC variant as the general clusterer. WADERs stays the pipeline reference (and is what A_GMX itself uses on "unmodelled" images); plain MFD+G stays the fastest choice when only isolated X-rays matter (Fe-55 calibration).

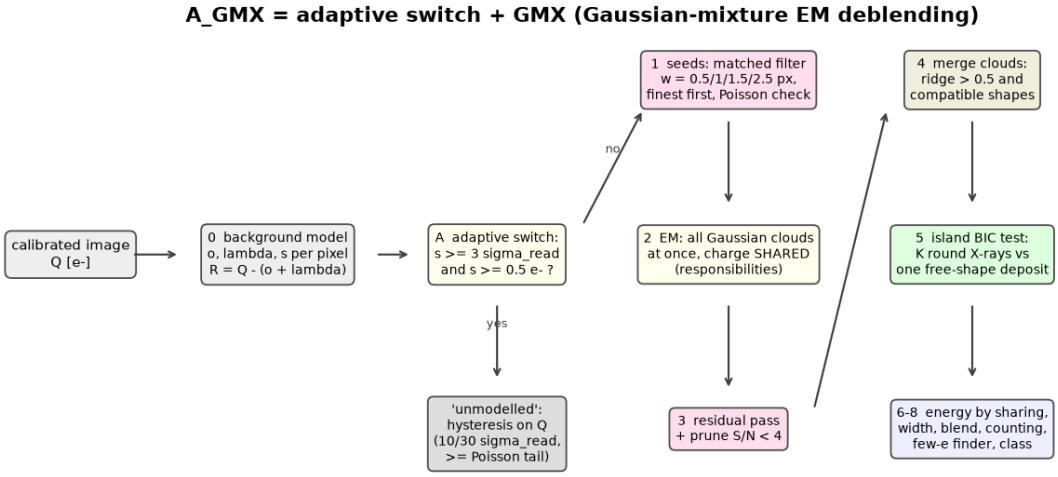


Figure 1: The steps of A_GMX (the numbers refer to the sections below).

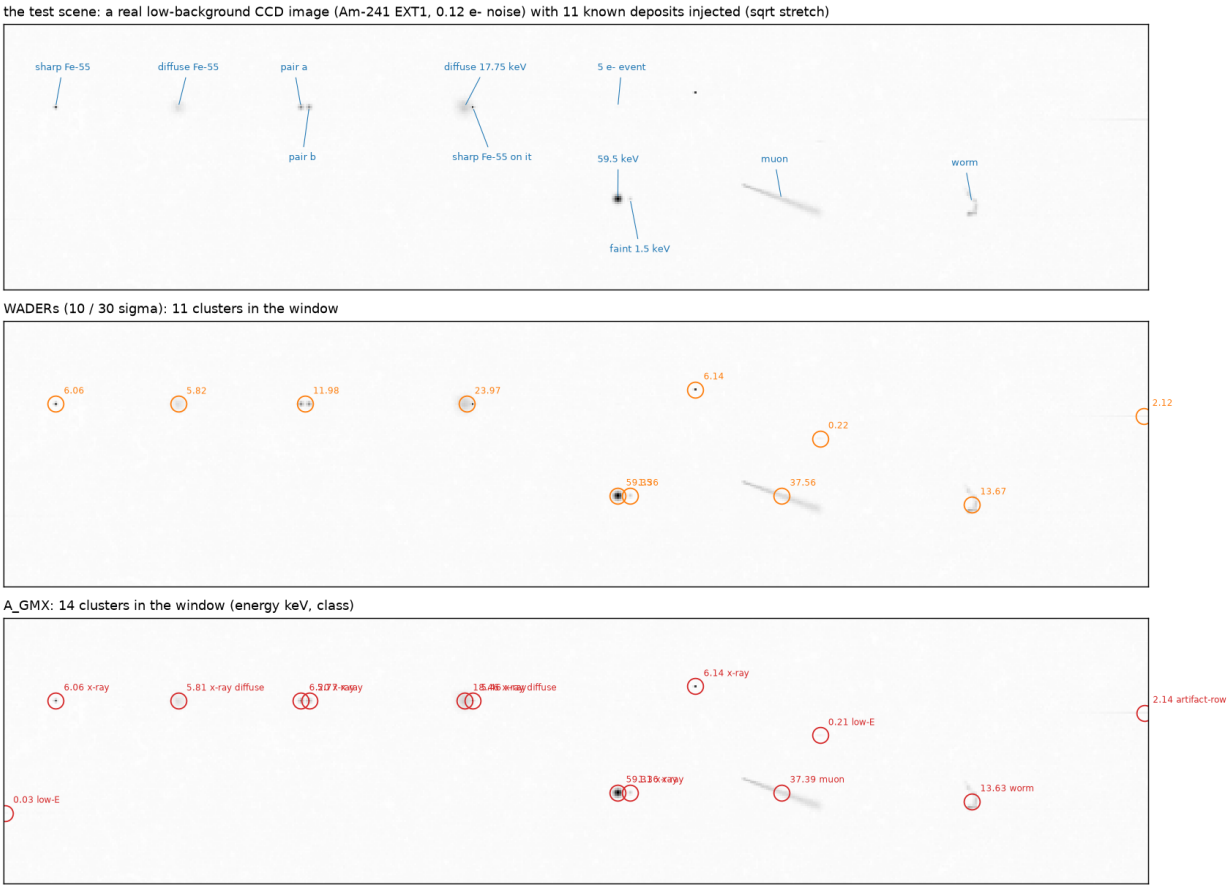


Figure 2: A real low-background CCD image (Am-241, EXT1, $0.12\,e^-$ read noise) with eleven known deposits injected (top). WADERs (middle) merges the 4-px pair into one 12-keV cluster and the sharp X-ray sitting on a diffuse one into one 24-keV cluster. A_GMX (bottom) separates both and labels the muon, the worm and the X-rays. Numbers: energies in keV.

2 The image model and notation

A calibrated image is a grid of pixel values Q_p in electrons (p = pixel). Each pixel holds

$$Q_p = o_p + n_p + \epsilon_p + D_p, \qquad n_p \sim \text{Poisson}(\lambda_p), \quad \epsilon_p \sim \mathcal{N}(0, s_p^2), \tag{1}$$

- o_p: offset (where "0 electrons" sits after calibration),
- n_p: dark charge, an integer number of electrons, Poisson with mean λ_p (e^-/pix),
- ϵ_p: read noise, Gaussian with width s_p (0.1–0.3 e^- on skipper CCDs),
- D_p: charge from particle deposits — what we want.

A deposit of charge A (electrons) whose charge cloud has covariance Σ (diffusion in the silicon: $\sim$0.3 px on the front side, up to $\sim$2 px on the back side) puts on average

$$D_p = A \int_{\text{pixel } p} \mathcal{N}(\mathbf{u}; \boldsymbol{\mu}, \Sigma)\, d^2u \tag{2}$$

electrons in pixel p. Energy = charge $\times$ 3.74 eV/e^-.

symbol	meaning
$R_p = Q_p - (o_p + \lambda_p)$	residual charge: $\approx 0\pm$ noise if empty, $\approx$ deposit charge otherwise
$V_p = \lambda_p + s_p^2$	variance of an empty pixel
$R_p^+ = \max(R_p, 0)$	the non-negative charge used as weight in the fit
$k = 1 \dots K$	clouds (components of the mixture)
$A_k,\ \boldsymbol{\mu}_k,\ \Sigma_k$	charge, centre, covariance of cloud k
$g_k(p)$	expected charge of cloud k in pixel p
$M_p = \sum_k g_k(p)$	model of the deposit charge in pixel p

3 Step 0 — the background model

Before looking for deposits, A_GMX describes the empty image (`agent_methods.background_map`). For every row, the pixels away from clusters are fitted with the **Poisson $\otimes$ Gaussian comb**

$$P(Q) = \sum_{n=0}^{\infty} \frac{\lambda^n e^{-\lambda}}{n!}\, \mathcal{N}\big(Q;\, o + n,\, s^2\big)\,, \tag{3}$$

i.e. integer electrons smeared by read noise (figure 3, bottom). The row values of o, λ, s are smoothed along the rows and in column blocks; pixels belonging to clusters are masked by a Poisson-tail test and the fit is iterated. This gives per-pixel maps o_p, λ_p, s_p (figure 3, top), hence

$$\text{mean}_p = o_p + \lambda_p, \qquad V_p = \lambda_p + s_p^2, \qquad R_p = Q_p - \text{mean}_p. \tag{4}$$

Hot columns. A column whose empty pixels are on average above the model by more than 5$\times$ the error of that mean (and by more than 0.05 e^-) gets that excess subtracted: $R_p \leftarrow R_p - \Delta_{\text{col}(p)}$.

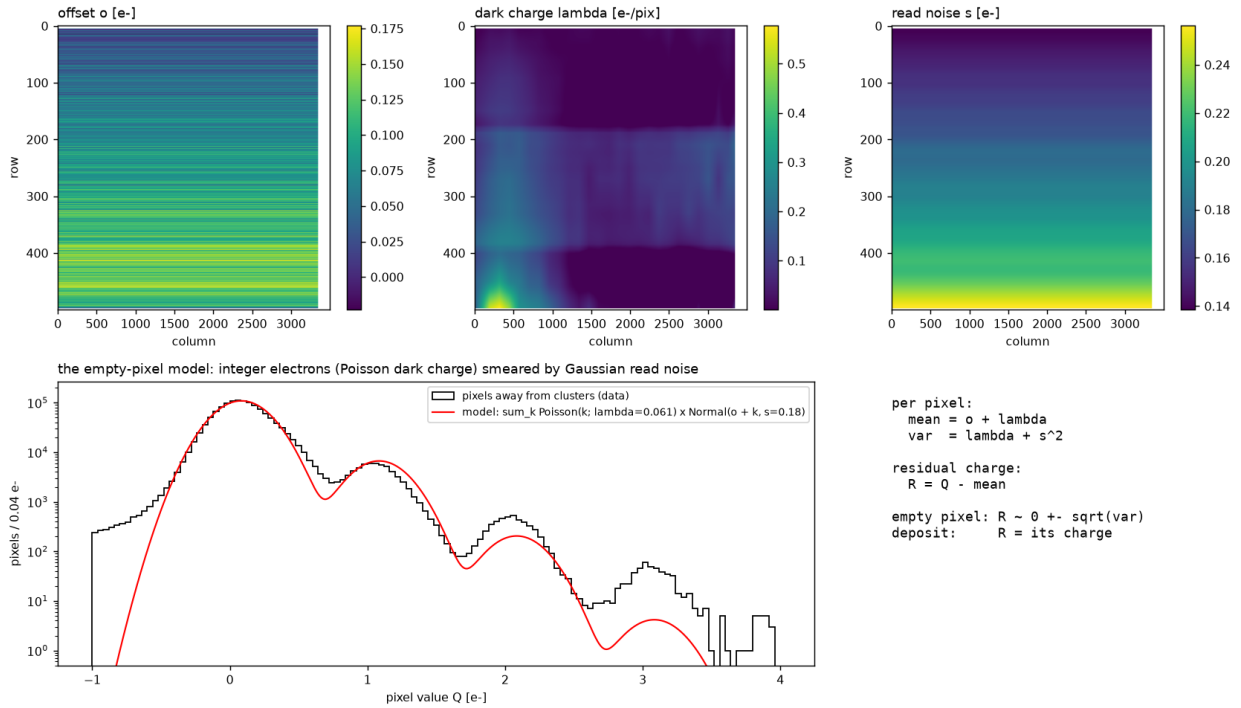


Figure 3: Top: the fitted offset, dark-charge and read-noise maps of the test image (dark charge grows along the readout). Bottom: pixels away from clusters (black) and the Poisson ⊗ Gaussian model (red): the peaks at 0, 1, 2 e^- are the integer electrons.

4 Step A — the adaptive switch

The background model can fail: on some images (here the PD06/DM12 Am-241 frames) the pixels scatter far more than dark charge + read noise allow — the fitted s is 1.5–3 e^- while the zero-electron peak is 0.37 e^- wide — and every model-based method then measures energies 5–10 % high. A_GMX therefore compares two estimates of the noise:

- **read noise from the zero peak:** $\sigma_{\text{read}} = 1.4826 \cdot \text{median}\,|Q - \text{median}(Q)|$ over the pixels with $|Q| < 0.5\,e^-$ (the median absolute deviation scaled to a Gaussian sigma; robust against pixels with charge);
- **fitted spread:** $s = \text{median}_p\, s_p$.

$$\boxed{s \geq 0.5\,e^- \quad \text{and} \quad s \geq 3\,\sigma_{\text{read}} \implies \text{“unmodelled”}} \tag{5}$$

Unmodelled → hysteresis clustering on the image itself (not on R, because the model's offset is not trusted either): pixels with $Q_p > \max(10\sigma_{\text{read}}, k-0.5)$ that touch (8 neighbours) form a cluster, kept if one pixel exceeds $\max(30\sigma_{\text{read}}, 9.5)\,e^-$; k is the smallest integer with $P(\text{Poisson}(\lambda) \geq k) \leq 0.10$; energy $= \sum Q_p$ over the cluster. This is WADERs with the thresholds bounded below by the dark-charge statistics.

Otherwise → GMX on the whole image (steps 1–8). In the wide test only the 27 PD06/DM12 image-CCDs of 878 were “unmodelled” (figure 4).

A_GMX also reports for every cluster the **event occupancy** of its 31 × 31-px neighbourhood: the fraction of pixels above $\max(1.5,\ 4\sigma_{\text{read}},\ k_{10^{-3}}(\lambda_p) - 0.5)$, i.e. holding *event* charge rather than dark charge (≥ 0.3 → labelled “crowded”). It is information only; GMX needs no crowded/sparse split (section 6.5 explains why).

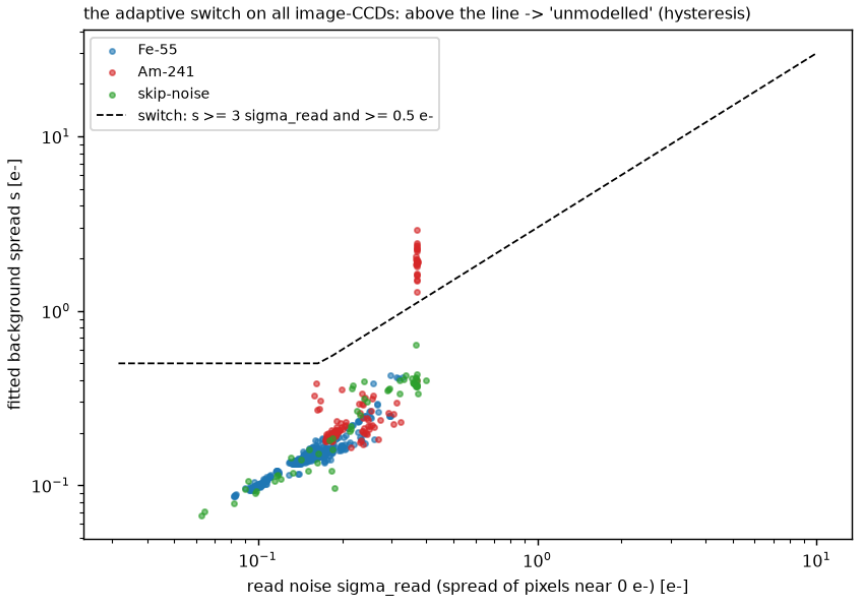


Figure 4: Fitted background spread s against the read noise σ_{read} for all 878 image-CCDs. Above the dashed line the background is not dark current + read noise ("unmodelled"): only the PD06/DM12 Am-241 images.

5 Step 1 — where are the deposits?

5.1 The matched filter

To decide whether there is a blob of width w centred at pixel c, a **matched filter** is used: a weighted sum of the residuals with the expected shape as weights. The weights are the fraction of a unit-charge Gaussian cloud of width w, centred on the centre of pixel c, that falls in each pixel $p = (i, j)$ — the exact pixel integral, separable into y and x:

$$h_w(p-c) = f(i-c_y)\, f(j-c_x), \qquad f(u) = \tfrac{1}{2}\Big[\operatorname{erf}\Big(\tfrac{u+\frac{1}{2}}{\sqrt{2}\,w}\Big) - \operatorname{erf}\Big(\tfrac{u-\frac{1}{2}}{\sqrt{2}\,w}\Big)\Big], \tag{6}$$

evaluated on a footprint of $\pm\lceil 4w+1\rceil$ px (`agent_methods._f1d`). With these weights,

$$S_w(c) = \sum_p h_w(p-c)\, R_p, \qquad \operatorname{Var}[S_w(c)] = \sum_p h_w(p-c)^2 V_p, \qquad \boxed{T_w(c) = \frac{\sum_p h_w(p-c)\, R_p}{\sqrt{\sum_p h_w(p-c)^2\, V_p}}} \tag{7}$$

T_w is the significance (in Gaussian sigmas) of a blob of width w at c. Summing many pixels with the right weights makes a deposit far more significant than any single pixel: a 5.9-keV X-ray spread over 20 pixels is invisible to a per-pixel 10σ threshold in its tails but has $T \gg 100$. Both sums are separable Gaussian convolutions (`gmx._sep_conv`), computed for four widths $w \in \{0.5, 1, 1.5, 2.5\}$ px (sharp front-side X-rays to diffuse back-side ones and blobs).

Normalisation and optimality. The overall scale of h_w cancels in T_w. Because the denominator is the exact standard deviation of the numerator for an empty region, T_w is a correctly normalised significance even where the noise varies from pixel to pixel. For noise that is the same in every pixel the filter is the optimal linear detector (largest signal-to-noise for a known shape; Turin 1960). Where V_p varies across the footprint (hot columns, steep dark-charge gradients) the optimal filter would also weight each pixel by $1/V_p$, $T_w^{\text{opt}} = \sum_p h_w R_p/V_p \,/\sqrt{\sum_p h_w^2/V_p}$; the unweighted numerator used here loses a little sensitivity there but keeps the false-alarm rate correct.

At very low dark charge the noise is Poisson rather than Gaussian, and the optimal Poisson template statistic is $\sum_p n_p \ln(1 + h_w A/B_p)$ (n_p the counts, A the deposit charge, B_p the background; Ofek the optimal Poisson template statistic is $\sum_p R_p \ln(1+h_w A/B_p)$ (Ofek & Zackay 2018) Zackay 2018); this is why every seed must also pass the exact Poisson box test below. The template sits on the pixel centre only (no sub-pixel offset): the EM fit of step 2 finds the centre.

5.2 Scale-space seeds

Seeds are local maxima (5×5 neighbourhood) with $T_w > 5$. The scales are visited **finest first**; a maximum at a coarser scale is kept only if no finer-scale seed lies within $\max(2, 1.5w)$ px; within a scale, seeds closer than that are suppressed (greedy, strongest first). Why: two X-rays 4–5 px apart give two maxima at $w = 0.5$ but one merged maximum in the middle at $w = 2.5$; finest-first keeps the two (scale-space detection, Lindeberg 1994). A lone diffuse X-ray has no fine-scale maximum (its peak is too flat), so its coarse-scale seed survives.

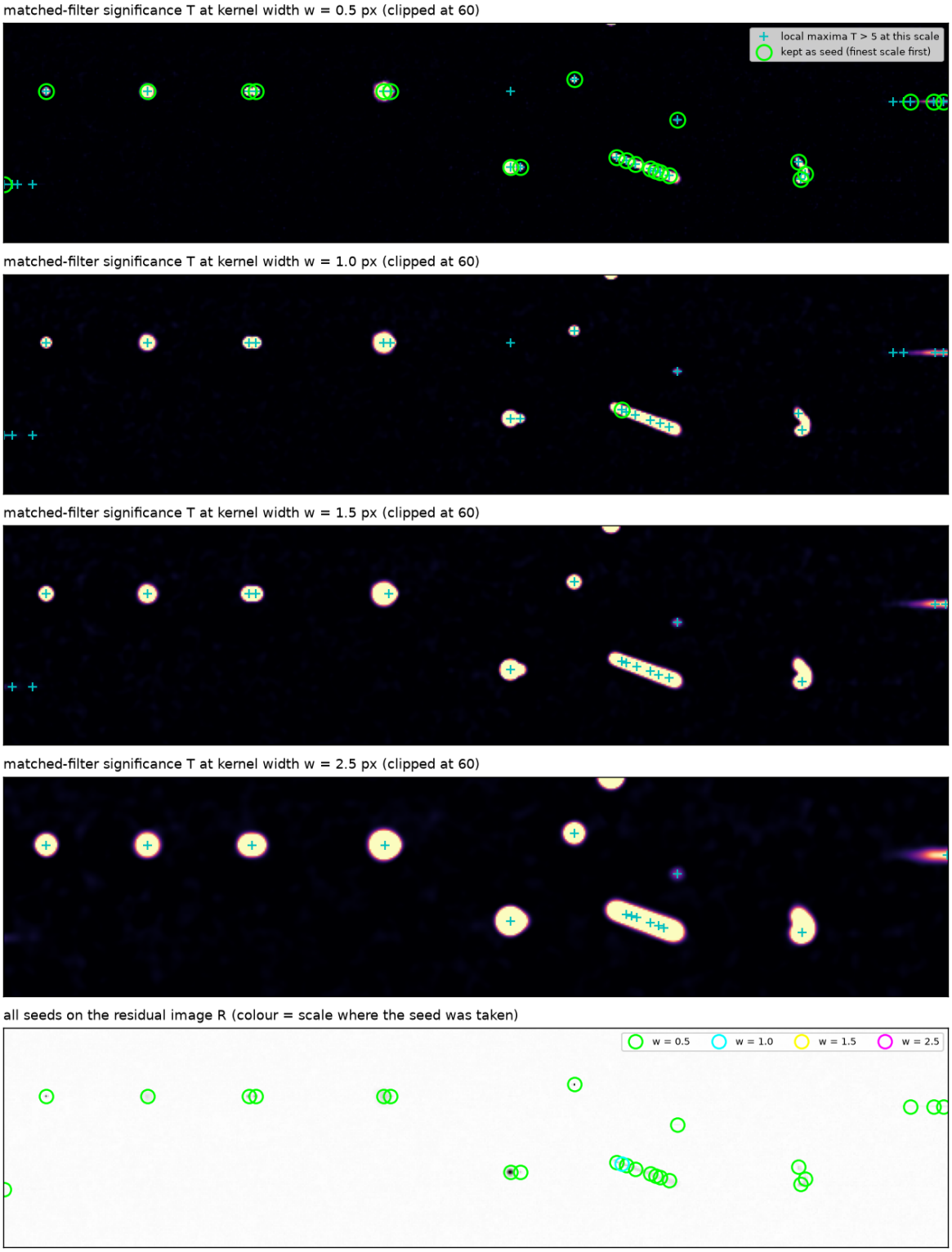


Figure 5: T_w at the four widths (clipped at 60) with the local maxima (+) and the seeds kept (circles). Bottom: all seeds on R, coloured by the scale they were taken at. The 4-px pair and the sharp X-ray on the diffuse one get two seeds each; the muon and the worm get several.

5.3 The Poisson check

T_w treats the noise as Gaussian, but dark charge is integer and Poisson: at low λ a lone box of 3 electrons is "6σ" in Gaussian terms but has probability $\sim 10^{-4}$ (figure 6). Every seed must therefore also pass an **exact counting test** on the 3×3 and the 5×5 box around it (the idea of

Li & Ma 1983 in γ-ray astronomy): with $S = \sum_{\text{box}}(Q_p - o_p)$, $\Lambda = \sum_{\text{box}} \lambda_p$, $\sigma_{\text{box}} = \sqrt{\sum_{\text{box}} s_p^2}$,

$$P(\text{Poisson}(\Lambda) \;\geq\; \lfloor S - 2\sigma_{\text{box}} + 0.5 \rfloor) < \Phi^c(5) = 2.9 \times 10^{-7} \tag{8}$$

(for either box). The $-2\sigma_{\text{box}}$ term protects against read noise pushing a box up by a fraction of an electron.

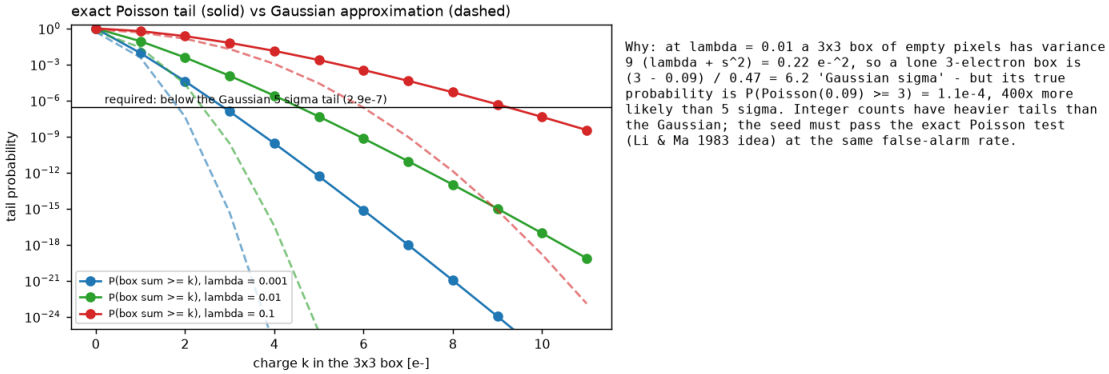


Figure 6: Tail probability of the 3×3-box charge from dark current alone: exact Poisson (solid) versus the Gaussian approximation (dashed). At low dark charge the Gaussian approximation is far too optimistic; the seed has to be below the horizontal line in the exact calculation.

Each surviving seed starts a round cloud: $\boldsymbol{\mu}$ = the seed pixel, $\Sigma = w^2\mathbf{1}$ (w = scale of the seed) and $A_0 = R_{\text{peak}} \cdot 2\pi(w^2 + \frac{1}{12})$ (the charge of a Gaussian with that peak height).

6 Step 2 — fitting all deposits at once

6.1 The model

$$M_p = \sum_{k=1}^{K} g_k(p), \qquad g_k(p) = A_k\, \mathcal{N}\big(\mathbf{x}_p;\ \boldsymbol{\mu}_k,\ \Sigma_k + \tfrac{1}{12}\mathbf{1}\big)\,. \tag{9}$$

Each cloud has its own charge A_k, centre $\boldsymbol{\mu}_k$ and **full** covariance $\Sigma_k = \begin{pmatrix} \sigma_{yy} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{xx} \end{pmatrix}$, so it can be round (an X-ray) or elliptical and tilted (a piece of a track). The $+\frac{1}{12}$ is the variance of a uniform distribution on a pixel: evaluating a Gaussian at the pixel centre with its variance increased by $\frac{1}{12}$ approximates the pixel integral (good for $\sigma \gtrsim 0.3$ px; for very sharp clusters the model's peak is too high — figure 9 shows it — but the energies do not use the model's peak, see section 10). **Since phase 2 (section 20) uncorrelated clouds ($|\rho| < 0.5$, every X-ray) use the exact pixel integral of their Gaussian (error functions) with a matching M-step; only strongly correlated track pieces keep the $+\frac{1}{12}$ approximation.** Each cloud is evaluated only within its 4σ ellipse (at least 2 px, at most 16 px), as sparse arrays.

6.2 What is maximised

Treat the deposit charge as **counts** with mean $M_p + \varepsilon_p$, where $\varepsilon_p = \sqrt{V_p}$ is a fixed "unexplained" level per pixel (a noise component of the mixture). The Poisson log-likelihood is

$$\ln L = \sum_p \Big[R_p^+ \ln\big(M_p + \varepsilon_p\big) - \big(M_p + \varepsilon_p\big) \Big]. \tag{10}$$

Maximising it over $\{A_k, \boldsymbol{\mu}_k, \Sigma_k\}$ is a missing-data problem: if we knew which cloud each electron came from, each cloud could be fitted alone. **Expectation–maximisation** (Dempster, Laird & Rubin 1977; for Poisson images Shepp & Vardi 1982, Richardson 1972, Lucy 1974) alternates two steps. For the amplitudes with the shapes held fixed, the update $A_k \leftarrow \sum_p r_{pk} R_p^+$ is the classic ML-EM (Richardson–Lucy) step and never decreases $\ln L$ (it assumes each cloud's profile sums to 1 over its footprint, true to $< 0.1\,\%$ for a 4σ footprint); the centre and covariance updates are the weighted-moment M-step of a Gaussian mixture. With the safeguards below (clipping, step limit) monotonic convergence is not guaranteed in general. Measured on the test image: after 30 iterations individual clouds inside a deposit still trade charge (30 more iterations change a cloud's A by a median 1.5 %), but the **cluster energies**, which share the measured charge, are stable: 60 instead of 30 iterations changes X-ray cluster energies by $< 0.004\,\%$ (99th percentile) and the median of all clusters above 0.5 keV by 10^{-6}; 2.6 % of clusters (extended or overlapping ones) change by more than 1 %.

6.3 E-step: who owns each pixel?

$$\boxed{r_{pk} = \frac{g_k(p)}{M_p + \varepsilon_p}} \qquad \left(\sum_k r_{pk} + \tfrac{\varepsilon_p}{M_p+\varepsilon_p} = 1\right) \tag{11}$$

r_{pk} is the expected fraction of pixel p's charge that came from cloud k (its **responsibility**). The noise share $\varepsilon_p/(M_p + \varepsilon_p)$ is ≈ 1 far from every cloud, so noise pixels are not handed to any cloud (figure 9, right, dotted).

6.4 M-step: update each cloud from its share

With weights $w_{pk} = r_{pk} R_p^+$ (the charge of pixel p attributed to cloud k):

$$A_k = \sum_p w_{pk}, \qquad \boldsymbol{\mu}_k = \frac{\sum_p w_{pk}\,\mathbf{x}_p}{A_k}, \qquad \Sigma_k = \frac{\sum_p w_{pk}(\mathbf{x}_p - \boldsymbol{\mu}_k)(\mathbf{x}_p - \boldsymbol{\mu}_k)^\top}{A_k} - \tfrac{1}{12}\mathbf{1}. \tag{12}$$

The charge of a cloud is the sum of its shares; its centre and covariance are the weighted moments of its shares (the $-\frac{1}{12}$ removes the pixel size again). Safeguards: variances clipped to $[0.02, 25]\,\mathrm{px}^2$, correlation $|\sigma_{xy}| \le 0.95\sqrt{\sigma_{yy}\sigma_{xx}}$, a centre moves at most 1.5 px per iteration, clouds with $A_k < 10^{-3}\,e^-$ are removed. 30 iterations.

6.5 Why this is the heart of A_GMX

No threshold decides which pixel belongs to which cluster. Where two deposits overlap, each pixel's charge is split in proportion to what each cloud predicts there. Three consequences:

1. **No percolation.** In a crowded field, threshold-and-connect methods chain neighbours into one blob (or need such high thresholds that tails are cut). Sharing has no such failure mode: this is why A_GMX beats WADERs on the crowded Am-241 CCD.

2. **No charge stealing.** A sharp X-ray on the shoulder of a diffuse one keeps only its own charge (figure 9).

3. **Tails are kept.** The faint tail of a diffuse cluster is explained by its Gaussian, not cut by a threshold: energies are unbiased at every width (figure 17).

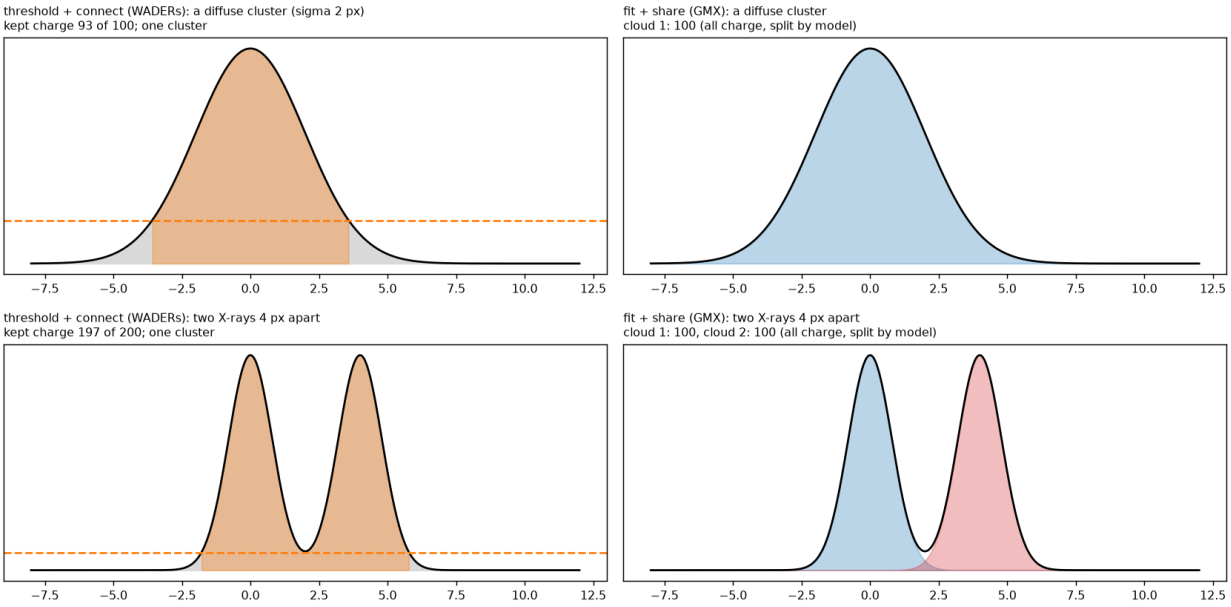


Figure 7: The difference in one dimension. Left, threshold + connect: the tails of a diffuse cluster below the threshold are lost (93 of 100 e^- kept here) and two X-rays 4 px apart are connected into one cluster. Right, fit + share: every electron is kept and the overlap region is split between the two clouds in proportion to their predicted charge.

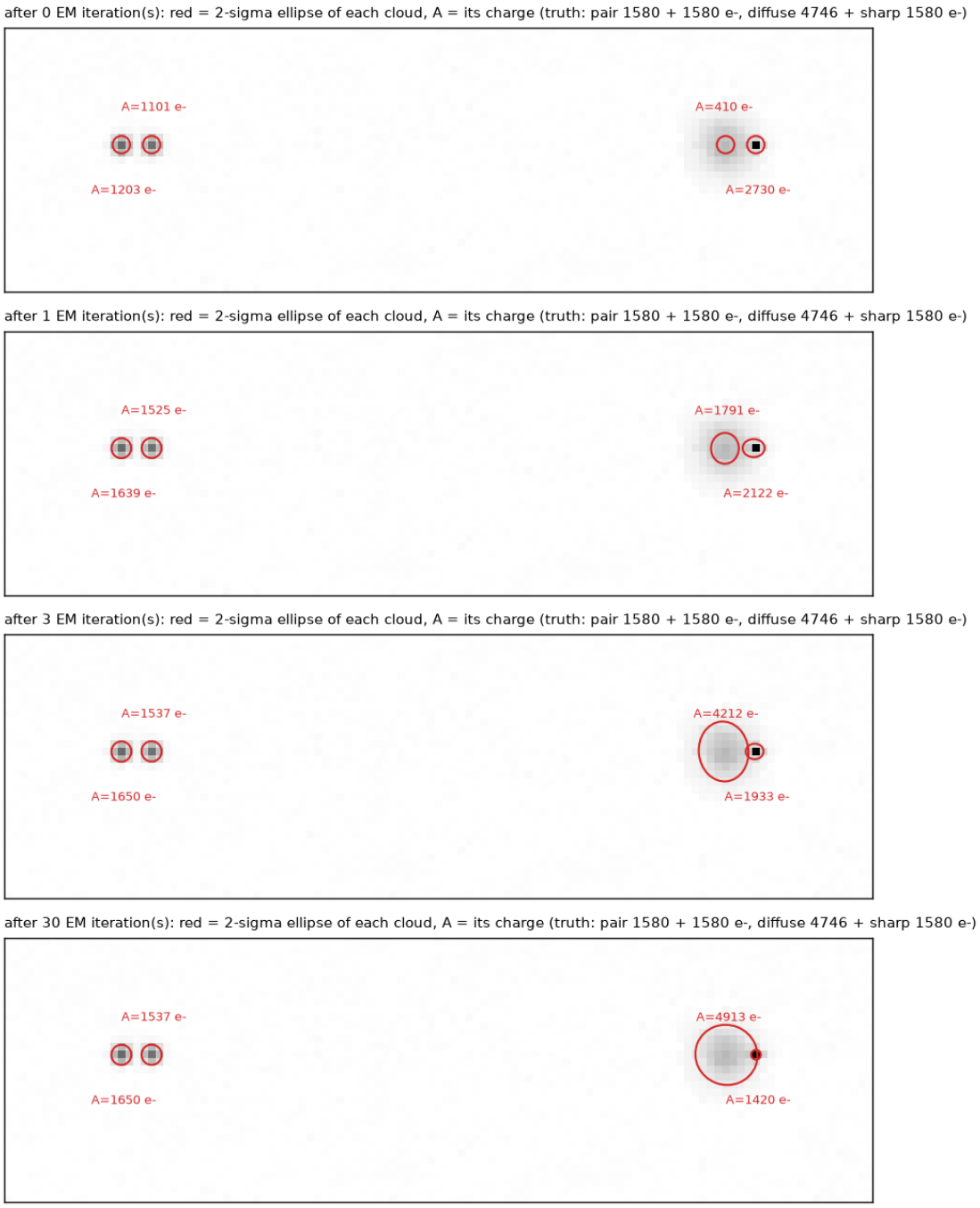


Figure 8: EM on the 4-px pair (left) and the sharp X-ray on a diffuse one (right): 2σ ellipses and charges A after 0, 1, 3 and 30 iterations. The diffuse cloud grows to its true width, the sharp one shrinks onto its pixel, and the charges converge to the truth ($1580 + 1580\,e^-$; $4746 + 1580\,e^-$).

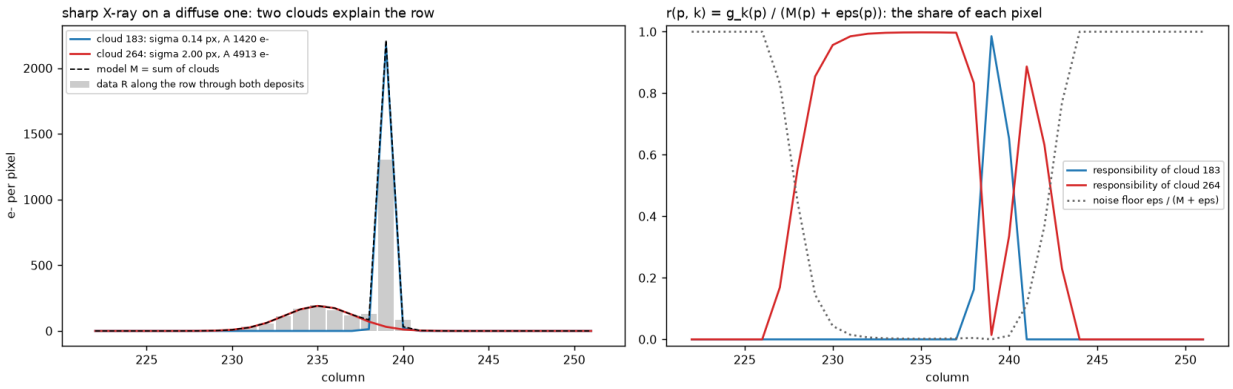


Figure 9: Left: the row through the sharp X-ray on the diffuse one — data (grey), the two clouds and their sum. Right: responsibilities; each pixel's charge is split between the two clouds, and far from both the noise share is ∼1.

7 Step 3 — residual pass and pruning

Residual pass. A faint deposit right next to a bright one can be hidden at the seed stage. After EM, the matched filter is run on the residual $R - M$ with the model's own Poisson variance added, $V'_p = V_p + M_p$, at the two finest scales; local maxima (3×3) with $T > 5$ more than 1 px from any cloud become new clouds, and EM runs 15 more iterations on all clouds (figure 10).

Pruning. The significance of cloud k is its charge divided by the uncertainty of that charge. For a linear fit of A_k with known profile $\pi_k(p) = g_k(p)/A_k$ and per-pixel variance v_p (read noise + dark charge + the other clouds' model, i.e. the variance if this cloud were absent), the Fisher information is $\sum_p \pi_k(p)^2/v_p$, so

$$\mathrm{S/N}_k = A_k \sqrt{\sum_p \frac{\pi_k(p)^2}{V_p + \big(M_p - g_k(p)\big)}}. \tag{13}$$

Clouds with $\mathrm{S/N} < 4$ are dropped; EM runs 5 more iterations.

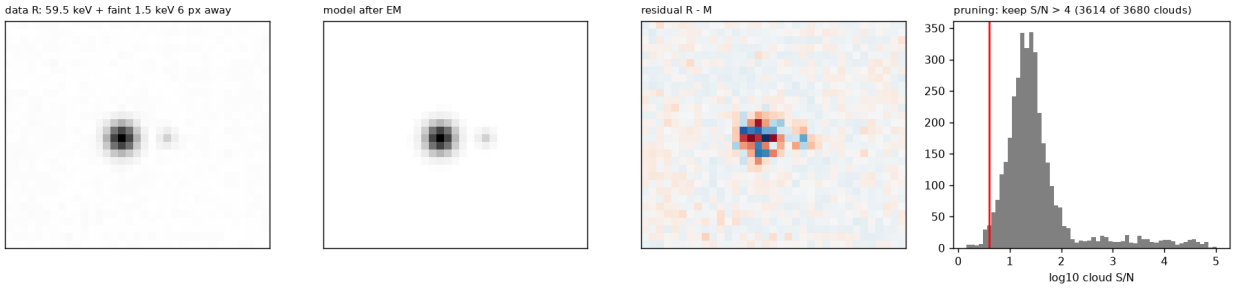


Figure 10: A 59.5-keV X-ray with a faint 1.5-keV X-ray 6 px away: data, model after the first EM, residual (the faint one stands out; circle = residual seed), and the S/N of all clouds of the image with the pruning cut.

8 Step 4 — which clouds are one deposit?

One deposit can need several clouds (a track needs many; a non-Gaussian X-ray may need two). Clouds are grouped by testing every pair closer than $3.5(\bar\sigma_i + \bar\sigma_j) + 1$ px ($\bar\sigma$ = geometric-mean width).

Ridge (dip) test. Walk the straight line between the two centres, $\mathbf{x}(t) = \boldsymbol{\mu}_i + t(\boldsymbol{\mu}_j - \boldsymbol{\mu}_i)$, $t \in [0, 1]$, and evaluate the two-cloud density $f(t) = g_i(\mathbf{x}(t)) + g_j(\mathbf{x}(t))$:

$$\rho_{ij} = \frac{\min_{t\in[0,1]} f(t)}{\min\big(f(0), f(1)\big)}. \tag{14}$$

$\rho \approx 1$: no valley, one connected object. $\rho \ll 1$: two separate peaks. Required: $\rho > 0.5$ (the modality criterion for merging mixture components; Ray & Lindsay 2005, Hennig 2010).

Shape test. Even without a dip, two clouds are merged only if **one** of these holds:

- similar widths: $\max(\bar{\sigma}_i, \bar{\sigma}_j)/\min(\bar{\sigma}_i, \bar{\sigma}_j) < 1.6$;
- a track: one cloud is elongated (axis ratio > 1.4) **and** its long axis points along the line joining them ($|\cos| > 0.8$);
- a shoulder: the narrower cloud is not a local maximum of f (it only shapes the flank of the broad one).

The case this protects: a **sharp X-ray on top of a diffuse one** is a maximum of its own with a very different width — kept apart. Pairs that pass are linked; the connected components of the links are the clusters.

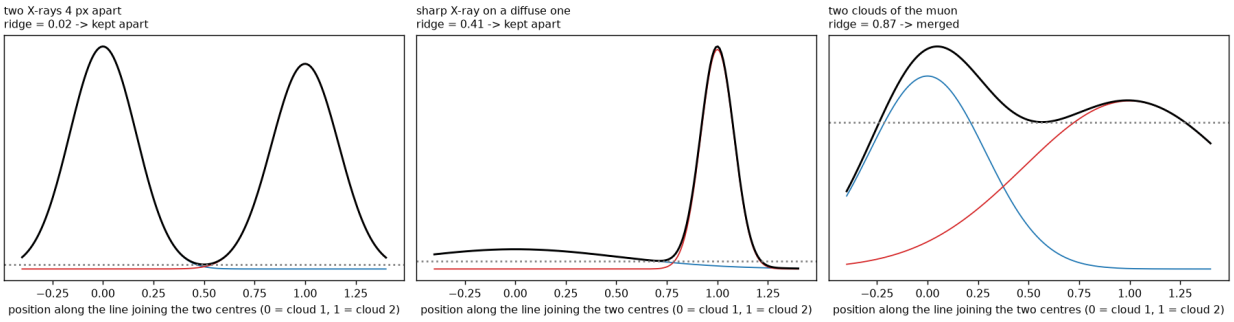


Figure 11: Two-cloud density along the line between centres. Left: two X-rays 4 px apart, deep valley ($\rho = 0.02$), kept apart. Middle: sharp X-ray on a diffuse one — no deep valley ($\rho = 0.41$) but the sharp one is a maximum of its own with a very different width, kept apart. Right: two clouds of the muon, $\rho = 0.87$, merged.

9 Step 5 — tracks versus several X-rays

Lumpy electron tracks ("worms") have real dips along them (their $\mathrm{d}E/\mathrm{d}x$ fluctuates), so the ridge test alone can split them. The **island test** asks the data directly.

Islands are connected regions (8 neighbours) of pixels above the Poisson-tail threshold of the dark charge at $\alpha = 0.01$. An island of at most 4000 px that holds $K \geq 2$ clusters (from step 4) is fitted with two hypotheses on its pixels (+2 rings):

- H_{xr}: K **separate X-rays** — one *round* Gaussian per cluster (started from the cluster's clouds, refitted by EM with $\Sigma = \sigma^2\mathbf{1}$): 4 parameters each (A, y, x, σ);
- H_{free}: **one irregular deposit** — the free-shape clouds already fitted: 6 parameters each $(A, y, x, \sigma_{yy}, \sigma_{xx}, \sigma_{xy})$, n_c clouds.

Each is scored with a weighted χ^2 and the **Bayesian information criterion** (Schwarz 1978), which charges $\ln n$ per parameter (n = pixels):

$$\chi^2_H = \sum_p \frac{\left(R_p - M_p^H\right)^2}{V_p + R_p^+ + (0.05\,R_p)^2}, \qquad \Delta\mathrm{BIC} = \left(\chi^2_{\mathrm{xr}} + 4K\ln n\right) - \left(\chi^2_{\mathrm{free}} + 6n_c\ln n\right). \tag{15}$$

The $(0.05R)^2$ term tolerates the fact that real X-rays are not perfect Gaussians (5 % model error). If $\Delta\mathrm{BIC} > 30$ — the free-shape description is so much better that it is worth its extra parameters — **the whole island becomes one deposit**. In crowded areas islands percolate beyond 4000 px and the test switches itself off (there, sharing does the work).

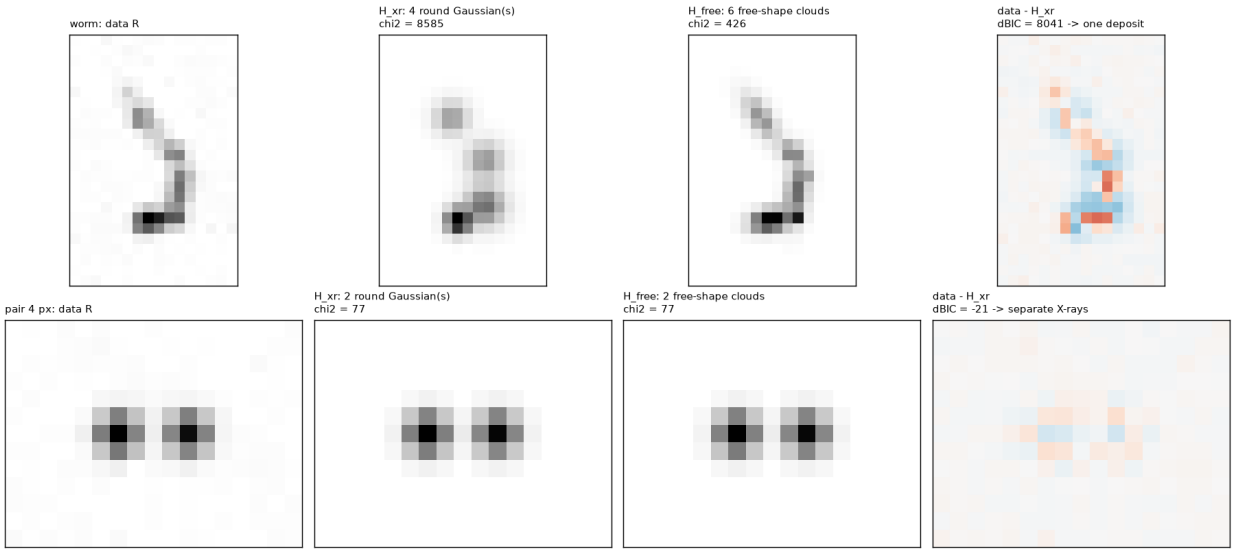


Figure 12: The island test on two islands of the test image. Top, the worm: 4 round Gaussians (one per cluster left by the ridge test) cannot follow the bent track (χ^2 8585, structured residual); 6 free-shape clouds can (χ^2 426); $\Delta\mathrm{BIC} = 8041 \rightarrow$ one deposit. Bottom, the 4-px pair: both descriptions fit equally well (χ^2 77), the round one has fewer parameters, $\Delta\mathrm{BIC} = -21 \rightarrow$ two separate X-rays.

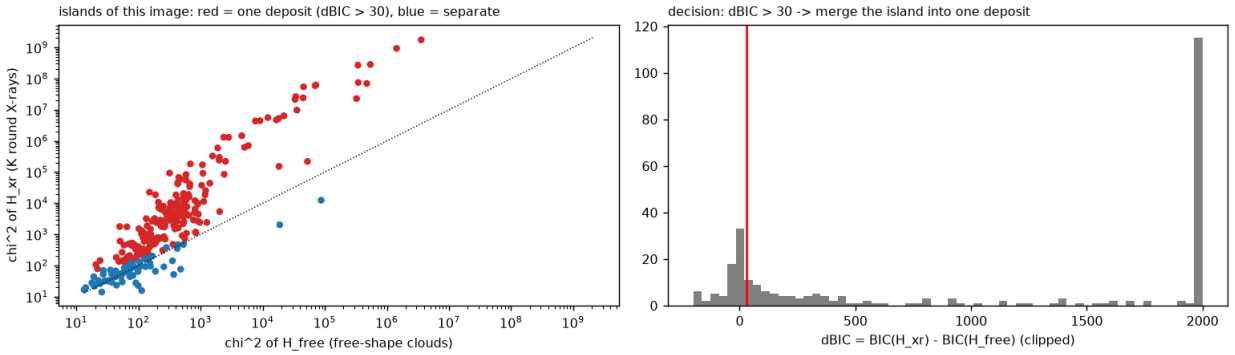


Figure 13: All islands of the test image: χ^2 of the two hypotheses (left) and $\Delta\mathrm{BIC}$ (right). Red: one deposit (tracks, worms, irregular blobs); blue: separate X-rays.

10 Step 6 — energy, shape, purity

Energy by sharing. The cluster responsibility of pixel p is $\tilde{r}_{pc} = \sum_{k \in c} g_k(p)/M_p$ (sum over the clouds of cluster c), and

$$\boxed{E_c = \sum_p \tilde{r}_{pc}\, R_p} \qquad [e^-;\ \times 3.74\ \text{eV/e}^-\ \text{for keV}] \tag{16}$$

The **measured** charge R_p is shared (not the model), so the energy does not depend on the Gaussian being exact; shared pixels are split, never counted twice.

- **Satellites.** A cluster with $E < 1\,\%$ of a neighbour's energy *and* $E < 1\,\text{keV}$ whose centre lies in that neighbour's domain or within 6σ (Mahalanobis) of one of its clouds is a tail fragment and is merged into it. (Until 25 October there was no 1-keV limit, and a real X-ray next to a very large deposit could be swallowed by it; see section 15.)
- **Growth.** Pixels above the Poisson-tail threshold ($\alpha = 0.01$) within 3 px of a covered pixel and connected to it, but covered by no cloud (CTI trails along the readout), are added to the nearest cluster.

Width. One-cloud clusters: $\sigma_{xy} = \sqrt{(\sigma_{yy} + \sigma_{xx})/2}$ of the fitted cloud; several clouds: second moment of the shared charge, $\sigma_{xy}^2 = \frac{1}{2}(\text{var}_y + \text{var}_x) - \frac{1}{12}$. Since the audit (D2) the second moment is reported for every cluster, and since phase 2 a single cloud with an exact-pixel EM width below 0.4 px reports the EM width (section 20).

Significance. $E_c \big/ \sqrt{\sum_p \tilde{r}_{pc}^2 V_p}$.

Purity: blend. How much of *other* clusters' predicted charge lies under this cluster's profile, relative to its own:

$$\boxed{\text{blend}_c = \frac{\sum_p M_p^{(c)} M_p^{(\neg c)}}{\sum_p \left(M_p^{(c)}\right)^2}}, \qquad M_p^{(c)} = \tilde{r}_{pc} M_p, \quad M_p^{(\neg c)} = M_p - M_p^{(c)}. \tag{17}$$

0 = isolated; 0.1 = 10 % of its profile is shared with neighbours. Cutting blend < 0.02 keeps clusters whose energy is not affected by sharing — on the crowded CCD this is what turns Np Lβ 15.5σ (WADERs) into 19.9σ (figure 20). The same idea is SExtractor's deblending flag, but here it is a number. **Edge** flags clusters touching the 2-px border of the active area.

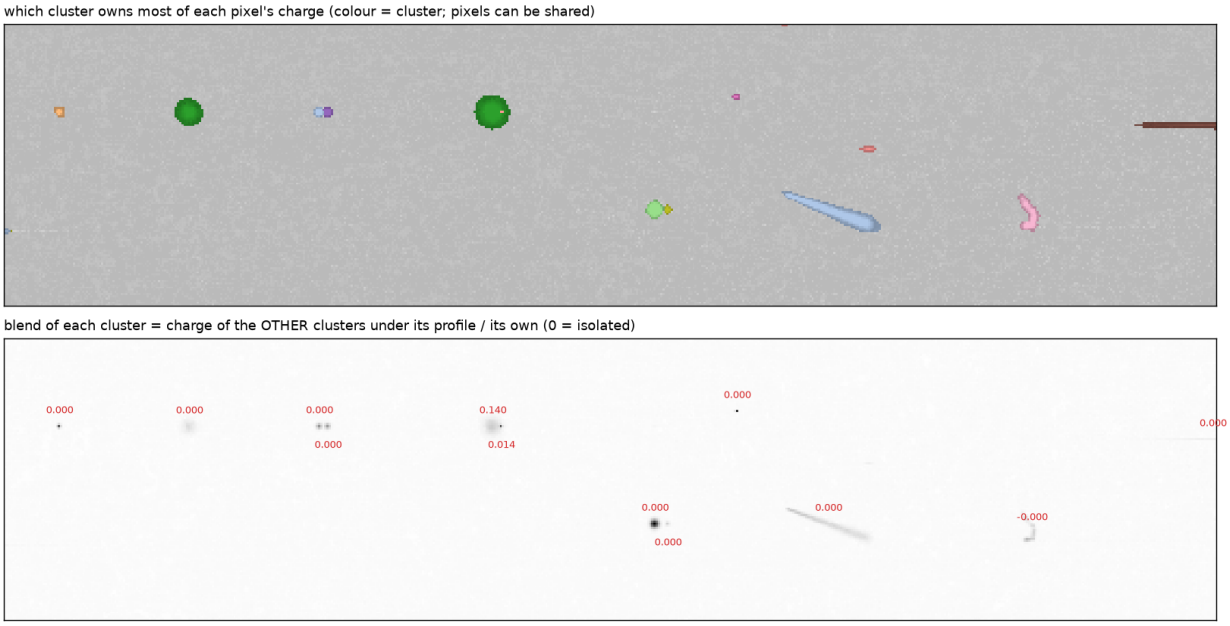


Figure 14: Top: which cluster owns most of each pixel's charge. Bottom: blend of each cluster (all ≈ 0 here except the sharp X-ray on the diffuse one, whose profile shares charge with its neighbour).

11 Step 7 — few-electron events

On photon-counting CCDs (median $s < 0.5\,e^-$) two extra rules apply.

Counting energies. For clusters below $100\,e^-$, the analog sum adds the read noise of every pixel ($\sqrt{N}\,s$, $0.5\,e^-$ for 13 pixels at $0.15\,e^-$), so the charge is often off by one. Instead each pixel the cluster mostly owns ($\tilde{r} > 0.5$) is **rounded** to whole electrons and the expected dark charge is subtracted:

$$E_c^{\text{count}} = \sum_{p:\ \tilde{r}_{pc}>0.5} \Big(\text{round}(Q_p - o_p) - \lambda_p\Big). \tag{18}$$

Counting finder. Groups of at most 9 connected pixels with $\text{round}(Q_p - o_p) \geq 1$ not covered by any cluster are added as "low-E" events when their charge is unlikely from dark current alone:

$$P\Big(\text{Poisson}\big(\textstyle\sum_{\text{group+1 ring}} \lambda_p\big) \geq S\Big) < 10^{-5}, \qquad S = \textstyle\sum_{\text{group}} \text{round}(Q_p - o_p). \tag{19}$$

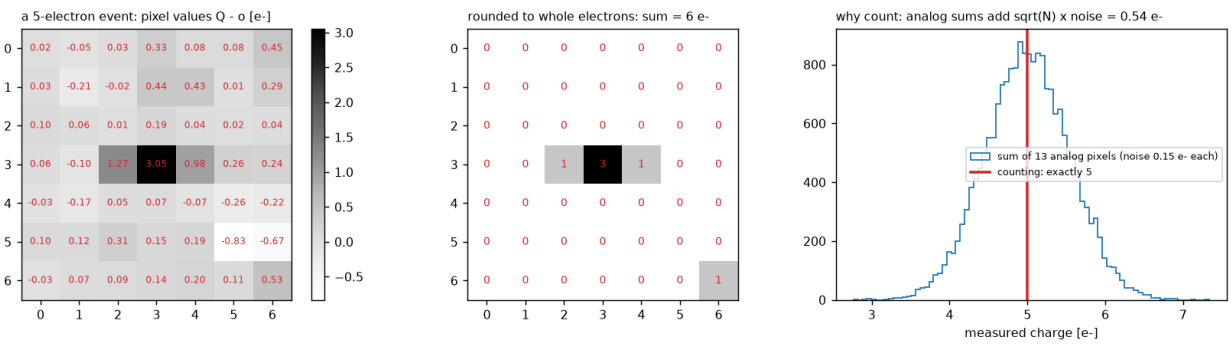


Figure 15: A 5-electron event: analog pixel values, the same rounded to whole electrons, and why counting is exact while an analog sum over 13 pixels spreads by $\pm 0.5\,e^-$.

12 Step 8 — classes

Every cluster gets a label from its shape (the rules of MHC): **low-E** ($E < 0.5$ keV); **x-ray** / **x-ray diffuse** (one round cloud or extent < 6 px; diffuse if $\sigma_{xy} \geq 1$ px); **alpha** (> 1 MeV); **artifact-row / -column** ($\geq 85\,\%$ of the charge in ≤ 2 rows / columns and a run of ≥ 8 px: serial-register streaks, hot columns); **muon** (long, straight skeleton: length ≥ 10 px, scatter of the cloud centres about a line < 1 px and $< 6\,\%$ of the length); **worm** (everything else extended). Labels are only checked on injected objects (alpha 0.94–1.00, x-ray 0.8, muon/worm swap $\sim$30 %).

13 The worked example

The test scene of figure 2: truth versus both methods (energies in keV).

deposit	truth	A_GMX	WADERs
sharp Fe-55	6.06	6.06 x-ray	6.06
diffuse Fe-55 (σ 1.6 px)	5.81	5.81 x-ray diffuse	5.82
pair, 4 px apart	6.20 + 5.77	**6.20 + 5.77**	11.98 (merged)
diffuse 17.75 keV + sharp Fe-55 3 px away	18.07 + 5.82	**18.46 + 5.46** (sum conserved)	23.97 (merged)
59.5 keV + faint 1.5 keV 6 px away	59.32 + 1.37	59.31 + 1.36	59.35 + 1.36
muon (40 px)	37.40	37.39 muon	37.56
worm	13.63	13.63 worm	13.67
5-electron event	0.019	not found	not found

The sharp-on-diffuse case is split, but 0.36 keV of the sharp X-ray's charge is assigned to the diffuse one: the Gaussian model of a very sharp cluster is approximate (section 6.1) and the sharing inherits that. The 5-e^- event sits on a CCD with $\sim$0.07 e^-/pix dark charge, where 5 e^- is at the detection limit (figure 19).

14 Performance against WADERs and the other methods

Five independent tests, all with the same injected objects for every method (same random seeds), plus real lines.

14.1 The wide test: every image on the machine

878 image-CCDs from 16 data sets (Fe-55 production modules DM01–DM28 and PD06–08; Am-241 AMTEst and PD06/DM12; the skip-noise study). Per image-CCD: 12 Fe-55 clusters (σ 0.3–1.6 px), 4 muons, 4 worms, 4 pairs 4 px apart. Success = one cluster with energy within 5 % (pairs: both, within 10 %).

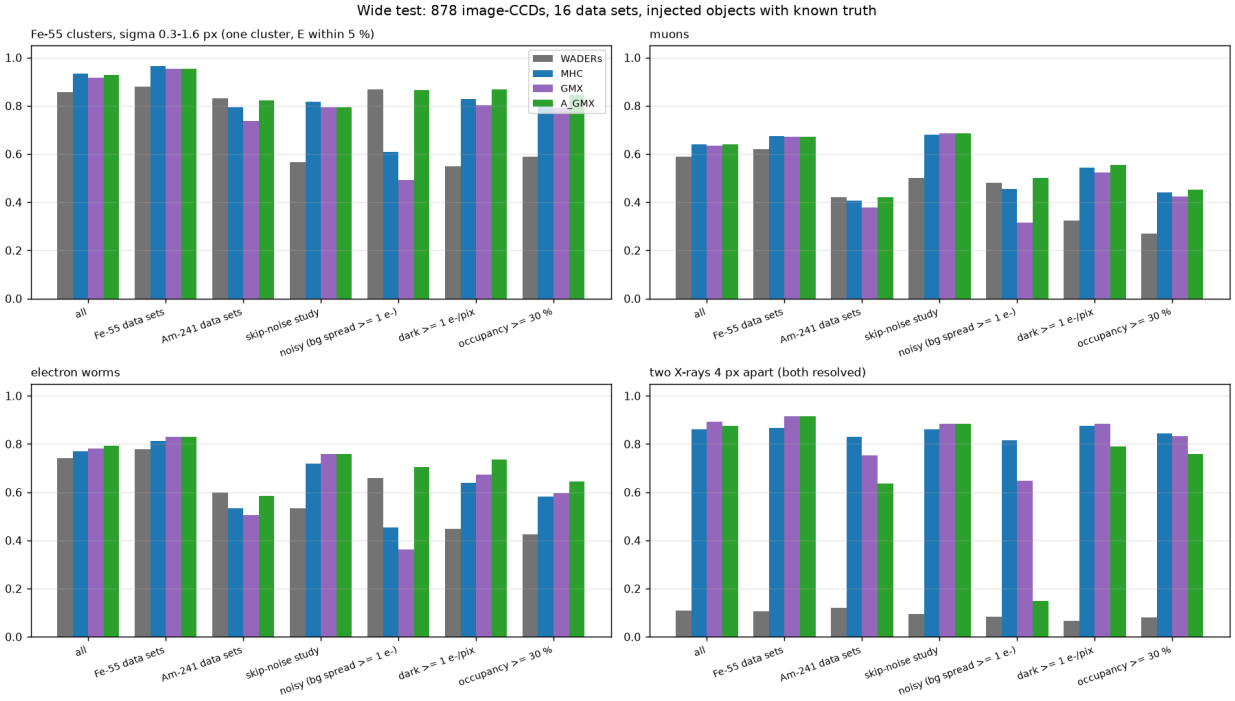


Figure 16: Success fractions by object type and image condition for WADERs, MHC, GMX and A_GMX.

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD
WADERs	0.856	0.52 %	0.59	0.74	0.11	0.01
B auto	0.905	0.59 %	0.62	0.78	0.13	0.01
MFD+G	0.933	0.21 %	0.18	0.37	0.94	0.14
hybrid v1	0.873	0.20 %	0.58	0.66	0.84	0.02
hybrid v2	0.930	0.21 %	0.63	0.68	0.85	0.05
MHC	0.933	0.21 %	0.64	0.77	0.86	1.61
adaptive (MHC engine)	0.931	0.21 %	0.64	0.77	0.83	1.26
GMX	0.916	0.19 %	0.63	0.78	0.89	0.69
A_GMX	**0.927**	**0.19 %**	**0.64**	**0.79**	**0.88**	0.69

Wide test, all 878 image-CCDs.

condition (Fe-55 success / worm / pair)	WADERs	MHC	GMX	A_GMX
Fe-55 data sets (705)	0.88 / 0.78 / 0.11	0.96 / 0.81 / 0.87	0.95 / 0.83 / 0.91	0.95 / 0.83 / 0.91
noisy, background spread $\geq 1\,e^-$ (27)	0.87 / 0.66 / 0.08	0.61 / 0.45 / 0.81	0.49 / 0.36 / 0.65	0.86 / 0.70 / 0.15
dark charge $\geq 1\,e^-$/pix (152)	0.55 / 0.45 / 0.07	0.83 / 0.64 / 0.88	0.80 / 0.67 / 0.88	0.87 / 0.74 / 0.79
event occupancy ≥ 30 % (194)	0.59 / 0.43 / 0.08	0.82 / 0.58 / 0.85	0.79 / 0.60 / 0.83	0.85 / 0.65 / 0.76

Wide test by image condition (number of image-CCDs in brackets).

14.2 Fe-55: energy versus cluster width, pairs, real Kα

DM09/DM10 (EXT1, EXT4, images 0–1; 220 injected clusters per width and CCD; pairs of σ 1.4-px clusters).

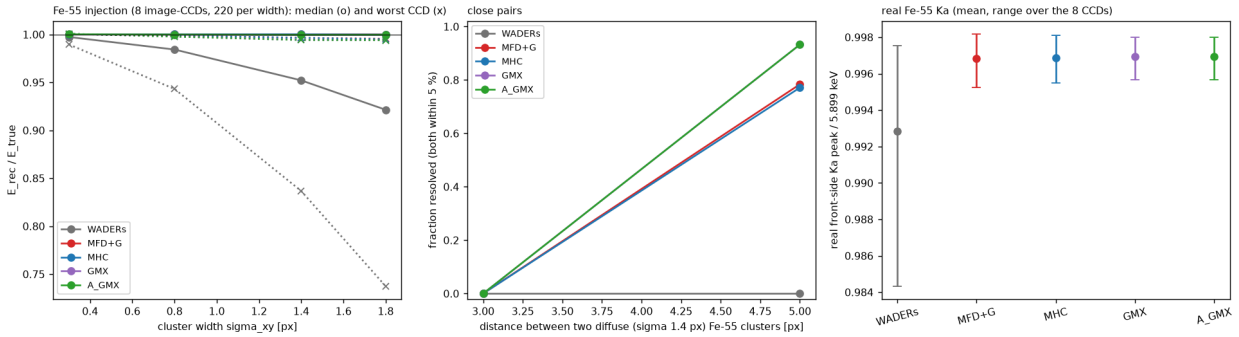


Figure 17: Left: median (circles) and worst-CCD (crosses) energy ratio against cluster width. Middle: diffuse pairs resolved. Right: real front-side Kα over the 8 CCDs.

method	σ 0.3	0.8	1.4	1.8 px	pair 5 px	real Kα/5.899	spread
WADERs	0.9974 (0.990)	0.9843 (0.943)	0.9521 (0.837)	0.9214 (0.737)	0.00	0.9928	1.32 %
MFD+G	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9999 (0.995)	0.78	0.9968	0.29 %
MHC	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9998 (0.995)	0.77	0.9969	0.26 %
A_GMX	1.0001 (1.001)	0.9999 (0.998)	0.9996 (0.994)	0.9997 (0.994)	**0.93**	0.9969	**0.23 %**

Median $E_{\mathrm{rec}}/E_{\mathrm{true}}$ (worst CCD in brackets) per cluster width; pairs resolved; real Kα.

14.3 Injected tracks, alphas and artifacts in real Am-241 images

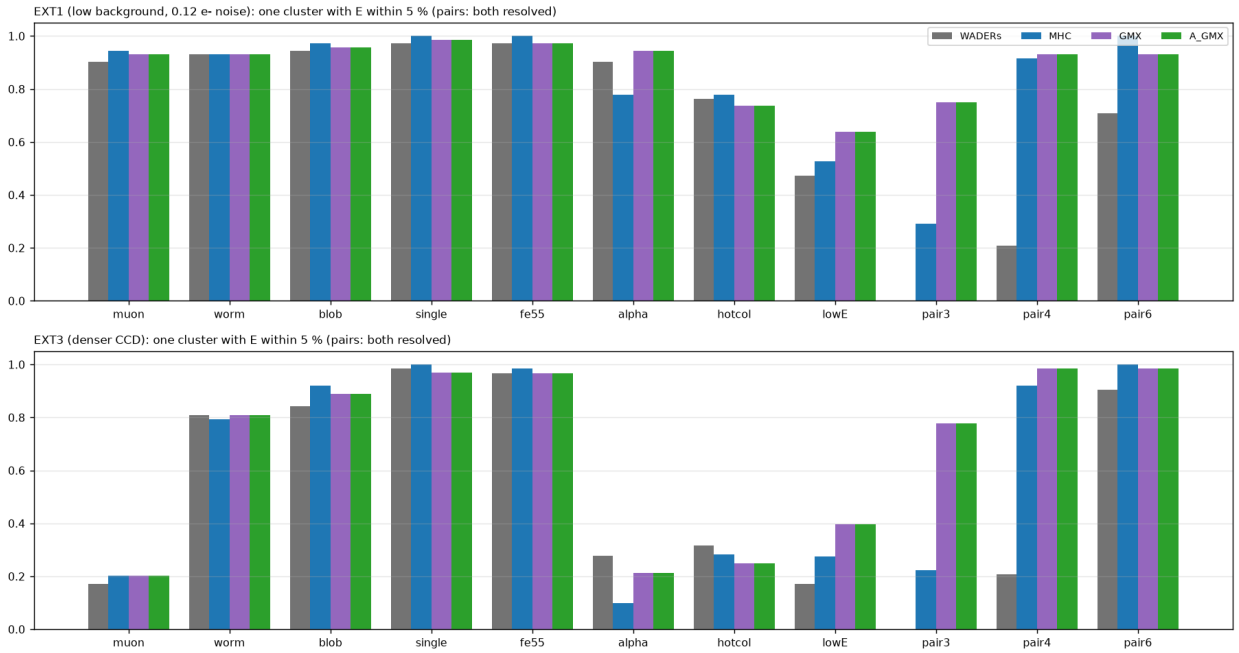


Figure 18: EXT1 (low background) and EXT3 (denser CCD): success per object type.

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.90	0.93	0.94	0.97	0.97	0.90	0.76	0.47	0.00	0.21	0.71
MFD+G	0.19	0.28	0.97	1.00	1.00	0.78	0.00	0.47	0.29	0.99	1.00
MHC	0.94	0.93	0.97	1.00	1.00	0.78	0.78	0.53	0.29	0.92	1.00
A_GMX	0.93	0.93	0.96	0.99	0.97	**0.94**	0.74	**0.64**	**0.75**	0.93	0.93

EXT1, 3 images × 3 seeds: success fraction per object type.

14.4 Few-electron events

Simulated blank images (Poisson dark charge + 0.15 e^- noise, 2 × 1.7 Mpix per dark-charge value), 800 events of 1–20 e^- in 1–5 pixels injected; fakes counted on a second blank image with nothing injected.

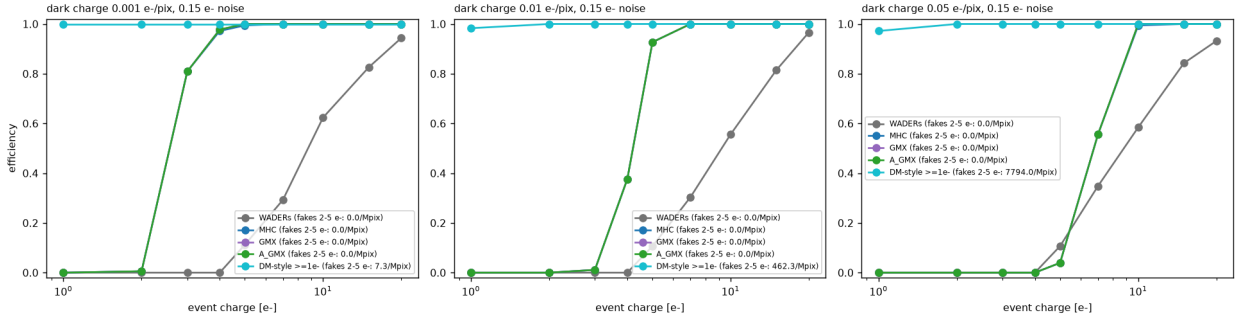


Figure 19: Efficiency against event charge at three dark-charge levels; the legend gives the fake clusters of 2–5 e^- per Mpix. A_GMX (= GMX = MHC here) finds 81 % of 3-e^- events at 0.001 e^-/pix and 93 % of 5-e^- events at 0.01 e^-/pix, with no fakes; at 0.05 e^-/pix it needs ∼7 e^- (and WADERs is slightly better at 5 e^- there: 0.11 vs 0.04 — the Poisson check is strict at that dark charge). WADERs needs $\gtrsim$ 10 e^- at low dark charge. Clustering of ≥ 1-e^- pixels (DM-style) finds everything, but its 2–5-e^- fakes from dark-current coincidences are 7 / 460 / 7800 per Mpix at 0.001 / 0.01 / 0.05 e^-/pix.

14.5 The crowded CCD: Am-241 10–30 keV constrained fit

29 images, EXT4 (in one corner every pixel holds > 20 e^-), Np L lines with fixed intensity ratios (Lépy et al. 2008), one energy scale k.

clustering (cut)	clusters	k	dev/ndf	Np Lα	Lβ	Lγ	γ 26.3	Ag K
WADERs ($\sigma_{xy} < 0.7$)	62 665	0.984	1.74	4.0	15.5	16.0	14.8	12.9
MFD+G ($\sigma_{xy} < 0.7$)	28 811	1.003	1.14	2.1	8.2	9.1	8.0	8.5
MHC ($\sigma_{xy} < 0.7$)	27 250	1.003	1.10	1.9	8.0	9.1	8.2	8.5
hybrid v1 ($\sigma_{xy} < 0.7$)	5 270	0.999	1.31	0.7	4.4	5.2	5.6	3.5
hybrid v2 ($\sigma_{xy} < 0.7$)	23 862	1.002	1.14	1.9	7.9	8.5	8.4	8.4
adaptive, MHC engine ($\sigma_{xy} < 0.7$)	60 158	0.986	1.51	3.0	14.4	15.4	14.0	12.0
A_GMX ($\sigma_{xy} < 0.7$)	230 633	0.996	3.34	1.6	8.5	13.9	17.1	9.1
A_GMX ($\sigma_{xy} < 0.7$, blend < 0.02)	82 073	**1.002**	1.93	3.7	**20.8**	**19.2**	**19.9**	**18.1**

Combined Np L + 26.3-keV significance Z: WADERs 27.1, A_GMX with the blend cut 34.8. Without the blend cut A_GMX keeps about twice as many clusters, because since the satellite limit (section 15) fragments above 1 keV in the crowded corner stay separate clusters instead of being absorbed by their big neighbour; they are exactly what the blend cut removes, and without it the lines are weaker (Z 23.7) and the scale low ($k = 0.996$). Checks: the same result in the even and the odd images separately (combined Z 24.1 and 24.7, k 1.001 and 1.002); an equivalent isolation

cut on WADERs clusters (nearest cluster >3–20 px) makes WADERs *worse* (Z 27.1 → 6.2), so the gain is the sharing, not the cut; A_GMX beats WADERs for every blend cut from 0.005 to 0.3 (Z 28.7–34.8; figure 20, right). Caveats: deviance/ndf 1.93 (vs 1.74) with structure at 25–28 keV, and σ_{xy} is a different width estimate from panaSKImg's fitted one (the 0.7 cut was not re-optimised).

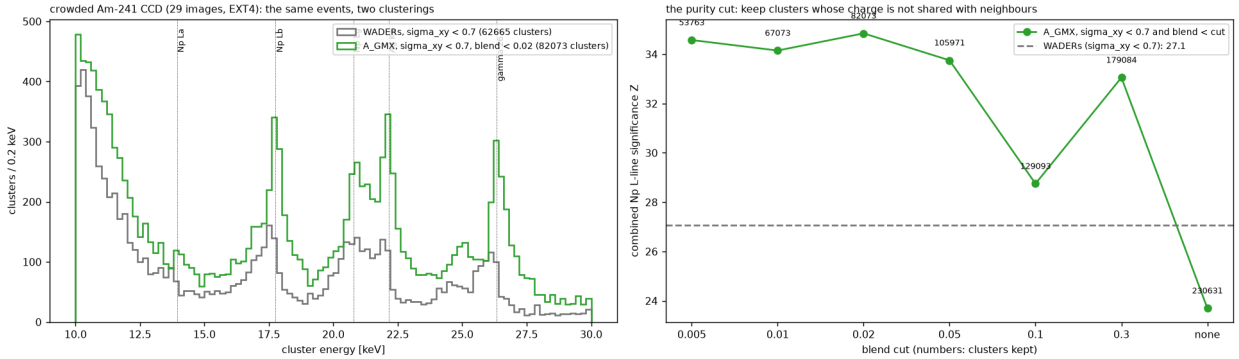


Figure 20: Left: 10–30-keV spectra of the crowded CCD, WADERs vs A_GMX with the blend cut — the same events, about twice the peak height above the continuum. Right: combined Np L significance against the blend cut.

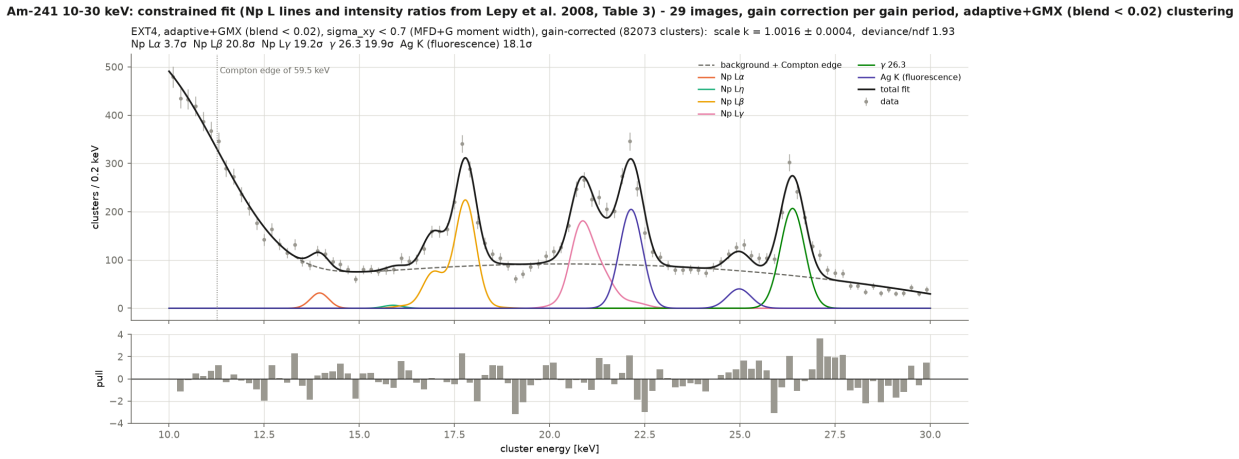


Figure 21: The constrained fit of the crowded CCD with A_GMX clusters ($\sigma_{xy} < 0.7$, blend < 0.02).

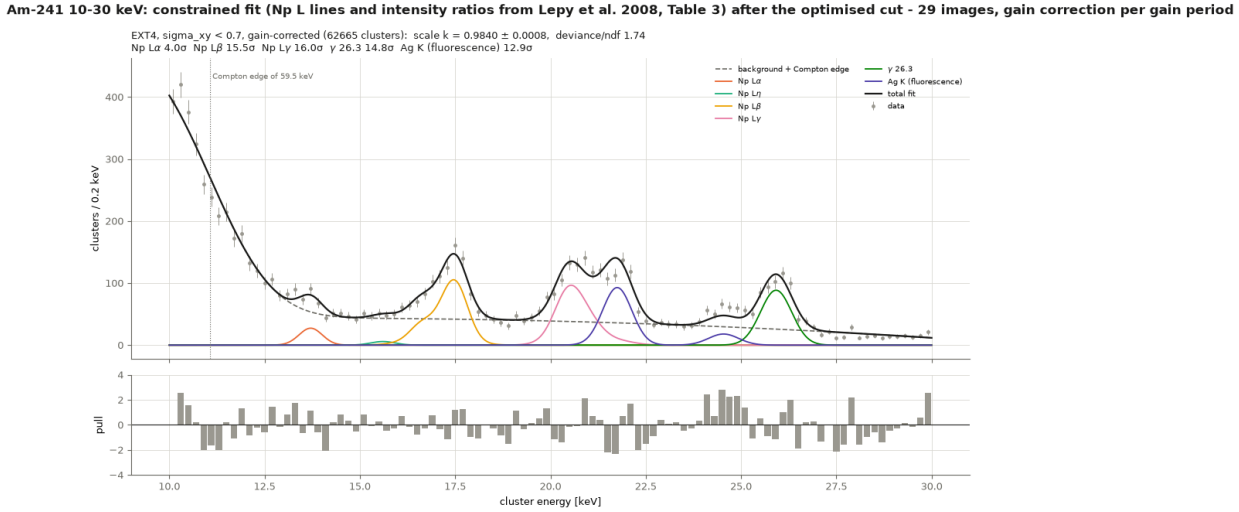


Figure 22: The same fit with the WADERs clusters ($\sigma_{xy} < 0.7$; panaSKImg's fitted width), for comparison.

14.6 Run time

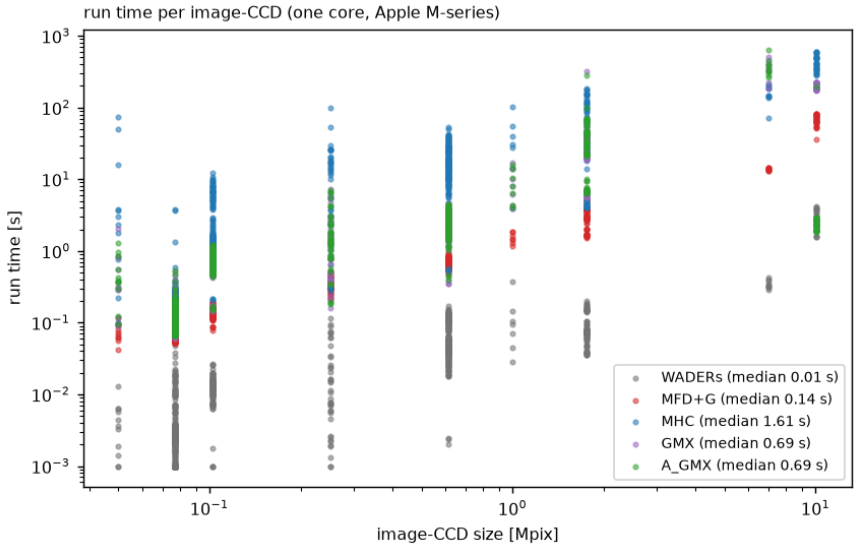


Figure 23: Run time per image-CCD against its size (one core; measured with 13 runs in parallel on 14 cores, so times are indicative). A_GMX: median 0.7 s; ∼1 s per Mpix on sparse images, minutes on the largest crowded CCDs.

15 Contamination: the high-side tail and the quality flags (25 October)

On the crowded Am-241 CCD the 59.5 keV line of A_GMX has a high-side tail and the lines sit slightly high ($k = 1.0014$; figure 24). This section shows where it comes from, what was tried, and what changed in the code.

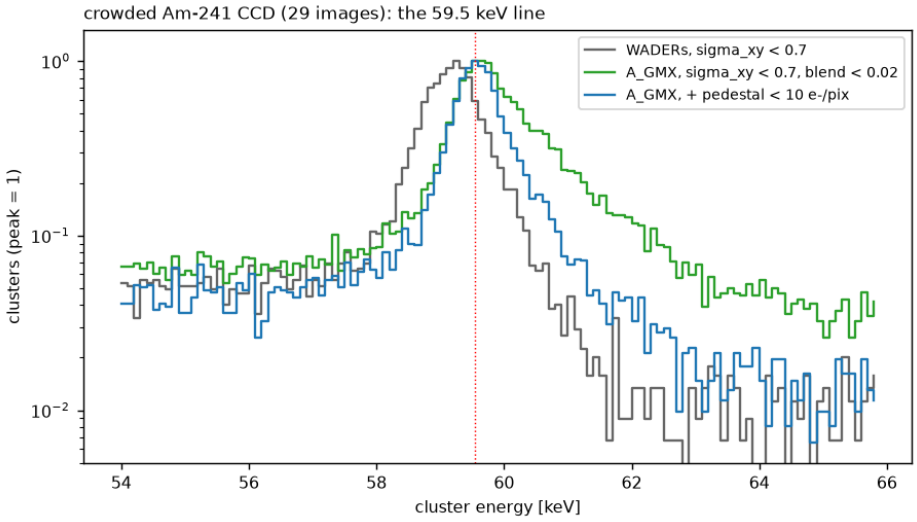


Figure 24: The 59.5 keV line of the crowded CCD (29 images): WADERs, A_GMX with the cuts of section 14, and A_GMX with the new pedestal cut. Scaled to the peak.

15.1 Is the algorithm biased? No.

Clusters of 5.9, 17.75, 26.34 and 59.54 keV were injected at *random* positions (also next to real events) into real images. Away from real events A_GMX is unbiased at every energy (median

$E_{\mathrm{rec}}/E_{\mathrm{true}} = 0.9998$, spread 0.03 % at 59.5 keV, no high tail), on the clean and the crowded CCD. Next to real events every method adds charge of the neighbour: median 1.033 (clean CCD) and 1.063 (crowded CCD) for A_GMX, similar for WADERs (figure 25).

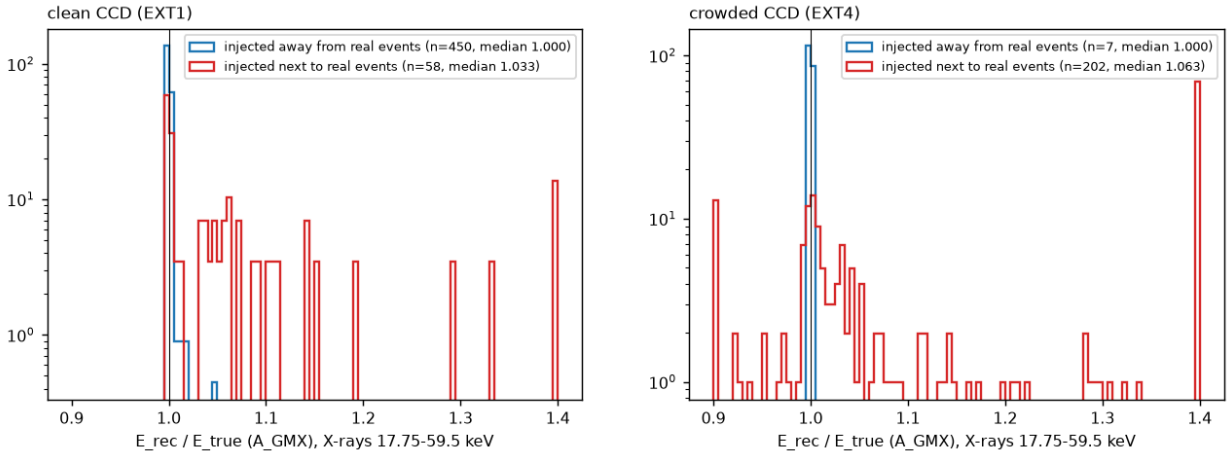


Figure 25: $E_{\mathrm{rec}}/E_{\mathrm{true}}$ of injected X-rays (17.75–59.5 keV) away from and next to real events.

15.2 Where the excess comes from

Two mechanisms, both real charge of other deposits:

1. **A charge sheet under the cluster** (dominant on the crowded CCD). Many small overlapping deposits make a diffuse sheet of 20–200 e^-/pix. Sharing gives the part of the sheet under a cluster's footprint to the cluster; the cluster still looks clean (one cloud, good χ^2, blend ≈ 0). The median charge in a ring 5–8 px around 59.5-keV-like clusters is 24 e^-/pix for clusters at the peak but 116–218 e^-/pix for clusters at 60.5–66 keV (figure 26). WADERs mostly avoids it because it only sums core pixels above $\sim$80 e^-.

2. **A neighbouring deposit merged into the cluster** (smaller effect on real data): such clusters have several clouds and a bad fit of their own model (median χ^2 15 vs 1; figure 27).

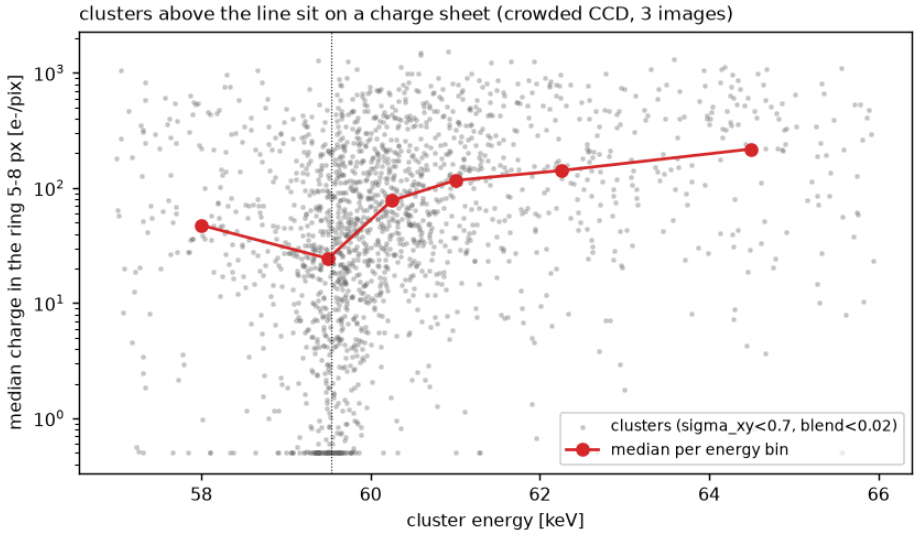


Figure 26: Charge in a ring 5–8 px around each 59.5-keV-like cluster against its energy: clusters above the line sit on a charge sheet.

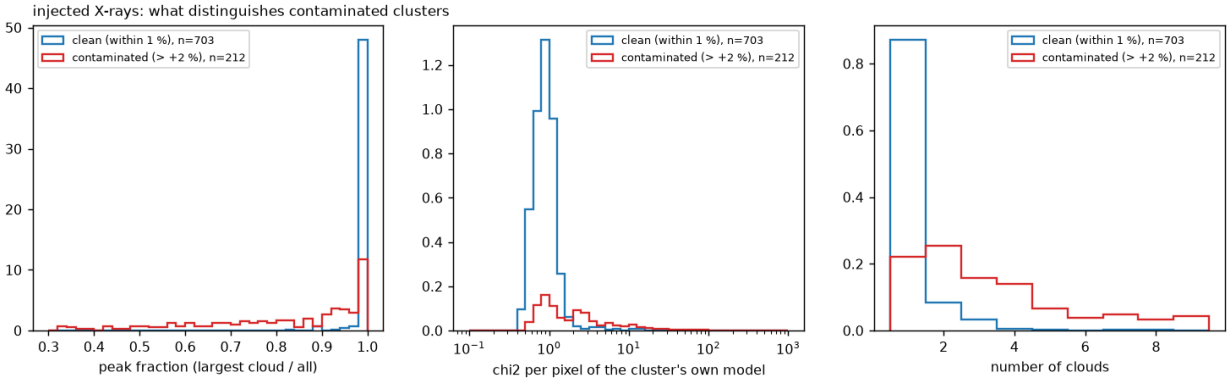


Figure 27: Injected X-rays: peak fraction, χ^2 of the cluster's own model and number of clouds for clean (within 1 %) and contaminated (> +2 %) clusters.

15.3 What does not work: re-measuring the energy

Growth off, pixel clipping at model +3–4σ, a local background from unexplained pixels, a core aperture (2 or 3 σ) with a model correction, a local pedestal from the ring (median or 25th percentile) subtracted over the footprint, and a joint fit of the cluster's shape + a flat level were all tested on the injections and the real lines. They either change nothing or add so much noise (the sheet is not uniform) that the lines broaden: e.g. subtracting the ring median removes the tail (0.60 → 0.22) but destroys the Np L fit (Z 34 → 5, $k = 0.965$).

15.4 What works: flags and a selection

Energies are left unchanged; every cluster now carries three quality flags (`gmx.general_cluster`):

$$\mathrm{ped}_c = P_{25}\{R_p :\ 5 \le |\mathbf{x}_p - \mathbf{c}| \le 8 \text{ px}\}, \qquad \mathrm{peakfrac}_c = \frac{\max_{k\in c} A_k}{\sum_{k\in c} A_k}, \tag{20}$$

$$\chi^2_c = \frac{1}{n_c} \sum_{p:\ \tilde{r}_{pc}>0.5} \frac{(R_p - M_p)^2}{V_p + M_p}, \tag{21}$$

the local pedestal (25th percentile of the residual charge in the ring; robust against neighbours in the ring), the share of the cluster's fitted charge in its largest cloud, and the fit quality of its own model. On injected X-rays, peakfrac > 0.95 keeps 98 % of clean clusters and rejects 67 % of contaminated ones; on the real crowded CCD the pedestal is what matters (figure 28):

selection (crowded CCD, 29 images)	clusters	59.5 tail/peak	k	dev/ndf	Z	Lβ / Lγ / γ26.3 / Ag K
WADERs ($\sigma_{xy} < 0.7$)	62 665	0.10	0.9840	1.74	27.1	15.5 / 16.0 / 14.8 / 12.9
A_GMX ($\sigma_{xy} < 0.7$, blend < 0.02)	82 073	0.62	1.0016	1.93	34.8	20.8 / 19.2 / 19.9 / 18.1
+ ped < 10 e^-/pix	48 839	0.23	0.9994	1.55	28.7	16.2 / 16.0 / 17.3 / 15.3
+ ped < 5 e^-/pix	41 344	0.18	0.9993	1.49	26.8	14.6 / 15.3 / 16.3 / 14.3

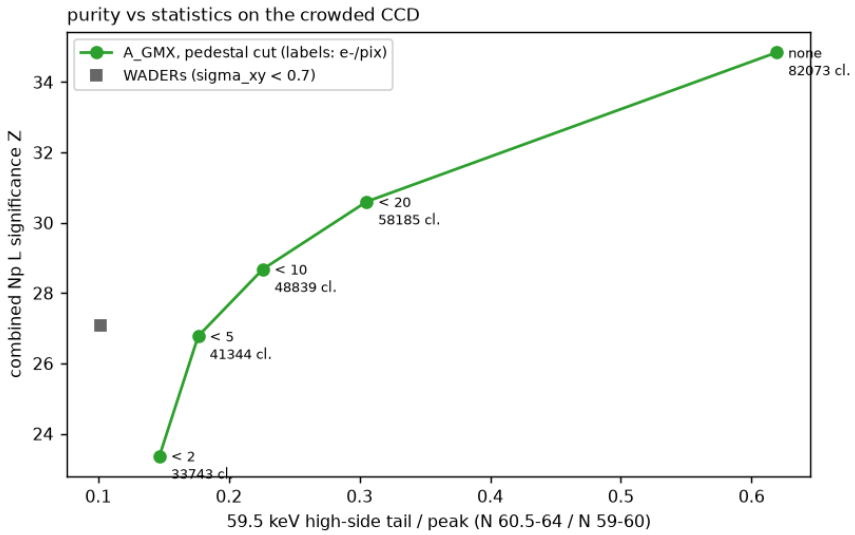


Figure 28: Purity against statistics: the 59.5 keV tail and the Np L significance for pedestal cuts of 20, 10, 5, 2 e^-/pix.

Satellites. Satellite merging absorbed any cluster with $< 1\,\%$ of a neighbour's energy, so a 5.9-keV X-ray next to a > 600-keV deposit was swallowed (crowded CCD, injected: median $E_{\mathrm{rec}}/E_{\mathrm{true}}$ 8.7). Only fragments below 1 keV are merged now (`sat_e_max`); the X-ray stays its own cluster (unit test `test_xray_next_to_big_deposit_not_merged`), with an energy a few % low when it sits on the big deposit's wing.

Recommended selection for spectroscopy. $\sigma_{xy} < 0.7$ (or the front-side cut of the analysis), blend < 0.02, ped $< 10\,e^-$/pix, not on the edge; add peakfrac > 0.95 where merged neighbours matter (crowded fields; not for Fe-55, see below). The flags are outputs: nothing is removed by `general_cluster` itself.

15.5 Effect on all X-ray lines and on Fe-55

Every Am-241 line fitted alone (Gaussian + linear background) on EXT1, EXT3 and EXT4 of the 29-image run, old and new code, with and without the flags (`am241_lines_flags.py`), and the front-side Fe-55 Kα of every data set of the wide test (`wide_agmx_flags.py`, `summ_wide_flags.py`):

CCD	selection	clusters	Np La	Np Lb	Np Lg	γ 26.3	γ 59.5	59.5 tail	k (EXT4)
EXT1	old A-GMX (sxy<0.7, blend<0.02)	13632	–	–	–	–	0.9867 (206)	0.23	
EXT1	new A-GMX (sxy<0.7, blend<0.02)	13715	–	–	–	0.9716 (251)	0.9867 (206)	0.23	
EXT1	new + ped<10	13715	–	–	–	0.9716 (251)	0.9867 (206)	0.23	
EXT1	new + peakfrac>0.95	13600	–	–	–	0.9723 (247)	0.9867 (208)	0.23	
EXT1	new + ped<10 + peakfrac>0.95	13600	–	–	–	0.9723 (247)	0.9867 (208)	0.23	
EXT3	old A-GMX (sxy<0.7, blend<0.02)	75580	–	0.9943 (200)	–	–	0.9989 (246)	0.11	
EXT3	new A-GMX (sxy<0.7, blend<0.02)	83037	–	–	–	–	0.9989 (246)	0.11	
EXT3	new + ped<10	81364	–	0.9953 (200)	–	–	0.9988 (242)	0.10	
EXT3	new + peakfrac>0.95	65083	–	0.9962 (200)	–	–	0.9994 (244)	0.14	
EXT3	new + ped<10 + peakfrac>0.95	63641	–	0.9962 (200)	–	–	0.9994 (239)	0.13	
EXT4	old A-GMX (sxy<0.7, blend<0.02)	65876	–	–	–	1.0038 (273)	1.0017 (353)	0.60	1.0014
EXT4	new A-GMX (sxy<0.7, blend<0.02)	82073	–	–	–	1.0041 (288)	1.0017 (352)	0.62	1.0016
EXT4	new + ped<10	48839	–	0.9977 (200)	–	1.0007 (245)	1.0006 (314)	0.23	0.9994
EXT4	new + peakfrac>0.95	70840	–	–	1.0053 (263)	1.0046 (296)	1.0025 (349)	0.69	1.0018
EXT4	new + ped<10 + peakfrac>0.95	41474	–	0.9983 (200)	–	1.0007 (250)	1.0012 (306)	0.24	0.9993

Every Am-241 line fitted alone: peak / true energy (width in eV); Np Lβ and Lγ are groups of lines, so their centres are not expected at exactly 17.75 / 20.78 keV (the constrained fit's k is the check there).

data set	n	Ka old (all)	Ka new (all)	front	front+ped	front+pf	front+both	front lost to ped	front lost to both
2025-08-27-DM07 DM09 DM10	102	5.8004	5.8000	5.7705 (63)	5.7705 (63)	5.7772 (62)	5.7772 (62)	0.0 %	28.7 %
2025-08-30-DM11 DM13 DM14	102	5.7960	5.7955	5.7732 (68)	5.7732 (68)	5.7915 (65)	5.7915 (65)	0.0 %	40.5 %
2025-09-03-DM15 DM16 DM17	102	5.7794	5.7789	5.7525 (67)	5.7518 (67)	5.7721 (65)	5.7721 (65)	0.1 %	37.4 %
2025-09-06-DM18 DM19 DM20	102	5.7719	5.7715	5.7387 (70)	5.7387 (70)	5.7524 (69)	5.7524 (69)	0.0 %	33.6 %
2025-09-13-DM24 DM25 DM26	90	5.9069	5.9070	5.8875 (65)	5.8875 (65)	5.8895 (66)	5.8895 (66)	0.0 %	38.0 %

Fe-55 K-alpha per data set: median fitted peak over image-CCDs [keV] (median width eV); old vs new A_GMX without cuts, new front side with the flag cuts. No K-alpha fit in any image-CCD of: 2025-09-07-DM18 DM19 DM20, 2025-09-09-DM21 DM22 DM23, 2025-09-16-DM08 DM11 DM12, 2025-09-19-DM07 DM27 DM28, 2025-10-31-DM04 DM05 DM06, 2025-11-04-DM10 DM24 DM25, 2025-11-08-PD06 PD07 PD08, 2025-11-13-DM01 DM02 DM03.

selection	image-CCDs	median K-alpha [keV]	spread [eV]	width [eV]
old A_GMX, all clusters	401	5.7934	66	79
new A_GMX, all clusters	401	5.7947	67	80
new, front	381	5.7709	69	67
new, front, ped<10	380	5.7704	69	67
new, front, peakfrac>0.95	343	5.7851	73	66
new, front, ped<10, peakfrac>0.95	343	5.7851	73	66

Fe-55 K-alpha, all Fe-55 image-CCDs: number fitted, median peak, robust spread over image-CCDs, median width. old = A_GMX before the 25 October changes, new = with them (satellite limit, flags, audit fixes).

data sets	image-CCDs	Fe-55	muon	worm	pair 4 px
Fe-55	705	0.951 / 0.954	0.672 / 0.672	0.829 / 0.829	0.912 / 0.915
Am-241	115	0.819 / 0.824	0.424 / 0.422	0.585 / 0.585	0.622 / 0.635
skip-noise	58	0.799 / 0.792	0.675 / 0.680	0.780 / 0.775	0.899 / 0.888

Injection test (same objects): success old / new A_GMX.

In short (all with the audited code): the changes leave Fe-55 alone — median Kα of all clusters 5.7934 → 5.7947 keV, per image-CCD the front-side Kα is unchanged (median difference 0.0 eV), and ped < 10 removes at most 0.1 % of the front-side Kα clusters (Fe-55 images have no charge sheet). peakfrac > 0.95 removes 28–41 % of them and moves Kα up by 14 eV, so it is *not* for Fe-55. On Am-241, EXT1 and EXT3 are unchanged (the EXT1 59.5-keV offset, 0.9867, is the fitted-gain calibration and is the same for every clustering); on the crowded EXT4 the pedestal cut brings every line to its energy: γ 26.3 1.0041 → 1.0007, γ 59.5 1.0017 → 1.0006, constrained-fit k 1.0016 → 0.9994, 59.5-keV tail 0.62 → 0.23. Injection success is unchanged within ±0.01.

16 Faster EM, few-electron finder and fairer tests (phase 1, 25 October)

Compiled EM. The EM iteration of section 6.5 is fused into one compiled pass (numba): the model M is accumulated first, then the responsibility-weighted moments, with the same footprints and the same M-step. The results are identical to 10^{-10} keV on a clean and on the crowded CCD; the crowded CCD takes 15 s instead of 32 s. `AGMX_NUMBA=0` selects the numpy version.

Few-electron finder with the full pixel likelihood. A group of touching pixels with $n_i \geq 1$ counted electrons is scored by

$$T = -\sum_{i \in g} \ln P(n_i \mid \lambda_i, s_i) - \ln N_{\text{shapes}}(m), \qquad P(n \mid \lambda, s) = \sum_k \text{Pois}(k; \lambda) \Big[\Phi\big(\tfrac{n+\frac{1}{2}-k}{s}\big) - \Phi\big(\tfrac{n-\frac{1}{2}-k}{s}\big) \Big], \tag{22}$$

the probability of the observed rounded counts under dark charge plus read noise (rounding miscounts included), with the number of 8-connected shapes of m pixels as the trial factor of the shape. The threshold is the value exceeded by the wanted number of fake groups (0.08 per Mpix by default, the earlier operating point) in blank images simulated from the image's own background; the analytic bound $\ln(N_{\text{pix}}/F)$ is used only when the blanks have too few groups. Section 18 gives the performance at the lowest dark currents.

Injection tests: objects away from real events. The injected objects were placed where their centre was quiet; a 15–35-px track could still cross a real event, and every method then "failed" on it. Objects are now placed only where their whole footprint (+2 px) avoids real events (`INJECT_FOOTPRINT=0` for the earlier placement, used for the wide test so that all methods stay comparable). On the injected tracks of the real Am-241 images (EXT3, the denser CCD) the muon success of every method goes from ≈ 0.2 to 1.00: the earlier numbers measured overlap, not clustering. With the new placement, A_GMX on EXT1: muon $0.93 \to 0.99$, worm $0.93 \to 0.99$, hot column $0.74 \to 0.90$, pairs 3/4/6 px 0.65/0.90/0.94 (new objects); on the dense EXT3 not every object kind finds a clean place any more (alphas, hot columns, low-E). On the 878-image wide test phase 1 changes no X-ray result (real Kα identical to 0.0 eV on every image-CCD, injections within 0.006).

17 Physics audit (25–26 October)

Every step of A_GMX and of the other methods was checked for physical and statistical soundness (simulations of skipper images with exactly known electrons; code review). The findings and what was done:

id	finding	fix	effect (simulation)
A1	a row counts as "comb resolved" when $\lvert\langle e^{2\pi i Q}\rangle\rvert > 0.018$, below the noise floor $\sqrt{\pi/4N}$ of pure noise: Gaussian noise of $1.5\,e^-$ was fitted as $s \approx 0.4\,e^-$, the switch did not fire	threshold $\max(0.018, 4/\sqrt{N})$; rows whose comb is washed out by event charge take o, λ, s from the nearest resolved rows (the read noise belongs to the amplifier); the pooled broad-noise fit only when $< 5\,\%$ of the rows are resolved	fakes on Gaussian noise 7966 → 1 per Mpix; crowded CCD unchanged
I1	the read-noise estimate (MAD of $\lvert Q\rvert < 0.5$) cannot exceed $1.4826 \times 0.25 = 0.37\,e^-$	robust spread of all active pixels when the first estimate is > 0.3	$0.37 \to 1.50$ / $3.00\,e^-$
A2	rounding counts noise $> +0.5\,e^-$ as an electron: λ high by $p = \Phi^c(0.5/s)$ (and the row mean, so every energy $-N_{\rm pix}p$)	$\lambda \leftarrow (\lambda - p)/(1 - 2p)$ on comb-fitted rows (offset unbiased)	$s = 0.3$, $\lambda = 0.001$: $0.051 \to 0.004$
—	λ clipped at $10^{-3}\,e^-$/pix (above DAMIC-M dark currents)	floor 10^{-6}; detection uses the measured single-electron occupancy (section 18)	
B1	bright hot columns hidden by the cluster mask (not flagged); flagged ones only had the mean subtracted (no variance)	column excess (clipped mean, no mask; revised against the neighbouring columns, section 20) added to the column's λ, so mean, variance and the Poisson tests follow	$+2\,e^-$/pix column: 50 → 0 fake few-electron clusters
C2	counting mode for $s < 0.5\,e^-$ although rounding miscounts are $\sim 10\,\%$ at $s = 0.4$	counting only for $s < 0.35\,e^-$	$s = 0.45$ blank: 313 → 0 fakes per Mpix
D1	the pixel model samples a Gaussian of variance $\sigma^2 + \frac{1}{12}$ at the pixel centre; its sum is 1.38 / 0.68 for $\sigma = 0.2$ (centred / half-pixel)	exact pixel integrals: phase 2 (section 20)	sharp X-rays fitted with one cloud
D2	the E-step floor $\varepsilon = \sqrt{V}$ pulls the fitted width low (1–4 % for Fe-55, up to 18 % for $200\,e^-$ at $1\,e^-$/pix): the width is the depth estimator	reported width = second moment of the cluster's shared charge (`sxy`); the EM width is kept as `sxy_fit`	Fe-55: 0.2–1.8 % instead of −1 to −4.5 %
G1	muon label from geometry only	a straight track is a muon only above 2.5 keV per projected pixel (a minimum ionising particle loses 0.27 keV/μm, $\geq$ 4 keV per 15-μm pixel)	physical muons 33/40 muon; the old too-faint test muons → "extended"
G2	saturated alphas not flagged	flag `saturated` (pixels at the image's maximum plateau)	

Physics-audit findings and fixes (`AGMX_PHYS_FIXES=0` restores the earlier behaviour).

Real data. 104 image-CCDs (every 12th Fe-55 image-CCD, every 6th Am-241 and all PD06/DM12), fixes on vs off with the same injected objects: Fe-55 data unchanged (injections equal to 0.004; front-side Kα −0.2 eV median, at most 7 eV). The noisy PD06/DM12 images (read noise ≈ 3.5–$4\,e^-$, now measured correctly) keep the WADERs-like branch with the earlier thresholds (pixel $> 4.5\,e^-$, seed $\approx 11\,e^-$): with the correct read noise the background model fits them and GMX would run there (pairs 0.15 → 0.97), but its footprint energy collects $\sqrt{N_{\rm pix}} \times 3.5\,e^-$ and the 59.5 keV line widens by $\sim 40\,\%$; choosing proper thresholds and energies for noisy images is an open item.

Found sound. The Poisson⊗Gaussian pixel model and its variance $\lambda + s^2$; the matched filter; the exact Poisson box check; charge conservation and unbiased energies (also at $1\,e^-$/pix and at every width); one dark-charge subtraction everywhere; units; the satellite limit, the width-ratio rule and the CTI growth.

Documented, not changed.

- Detection fakes depend on the dark charge: none in 4 Mpix at $\lambda \leq 0.05$, about one 16–27-e^- cluster per 3–4 Mpix at $\lambda = 1$–$2\,e^-$/pix.
- Footprint noise: $\sigma_E^2 = \sum_p \tilde{r}_p^2 V_p$ over the 4σ ellipse; Kα ($\sigma = 1.5$ px): $2.2\,e^-$ at $\lambda = 0.01$, $11\,e^-$ at $\lambda = 1$, comparable with the Fano width $13.7\,e^-$. A threshold sum has less noise but loses the tails.

- The ridge test merges equal clouds up to $d = 3.3\,\sigma_{\mathrm{tot}}$ although they are bimodal above $2\,\sigma_{\mathrm{tot}}$ (≈ 3.5 px back side, 1.4 px front side); only `peakfrac` shows such merges.
- The island BIC test uses a heuristic χ^2 likelihood with an un-derived margin of 30, compares a global model with an island refit (biased toward merging), and its round-X-ray hypothesis penalises elliptical X-rays (CTI, column binning). To be recalibrated on injected tracks before changing it.
- The blend cut removes deep (diffuse) events preferentially in crowded fields: fine for line positions, an efficiency correction for rates.
- In the WADERs-format files, A_GMX Energy is dark-subtracted (WADERs includes $N_{\mathrm{pix}}\lambda$), and `STD_X/Y`, `DX/DY`, `npix` are footprint sizes: WADERs cuts on them do not transfer (`fSTD` does).
- Test harness: the simulated muons deposited $250\,e^-$/px (0.94 keV/px), four times below a minimum-ionising particle.

18 Few-electron events at the lowest dark current (26 October)

For rare-event searches the decisive regime is a dark current of 10^{-4}–$10^{-3}\,e^-$/pix. Three problems limited A_GMX there: the λ floor of 10^{-3}; the analytic threshold, which counts one trial per pixel although at such dark currents an image has only a handful of candidate groups ($0.001\,e^-$/pix: 0.000 fakes per image instead of the wanted 0.14, and 3-e^- efficiency 0.77 instead of 0.93); and the fitted λ map, which fluctuates far below and above the truth when a fit cell holds only a few dark electrons. The detection background is now the dark charge implied by the measured rate of *isolated* single-electron pixels (dark electrons are isolated; real events come with neighbours), smoothed over ±30 rows, with the rounding miscounts of the read noise; A_GMX's per-event finder uses it conservatively (never below the fitted map), the candidate catalogue (section 19) exactly.

dark charge [e^-/pix]	$3\,e^-$ found (before → now)	$4\,e^-$ found	fakes per 1.7 Mpix (before → now)
0.0001	0.91 → **0.96**	1.00	0.00 → 0.30
0.0005	0.90 → **0.93**	0.99	0.00 → 0.37
0.001	0.84 → 0.85	1.00	0.07 → 0.07
0.003	0.39 → 0.40	0.98 → 0.96	0.30 → **0.07**
0.01	0.04	0.37 → **0.53**	0.43 → **0.23**

Full A_GMX on simulated images ($0.15\,e^-$ read noise): 1800 events of 2–$5\,e^-$ (Gaussian σ 0.2–0.6 px) and 30 blank images per dark charge.

What a per-event decision cannot do at the lowest dark current: the remaining fakes there are mostly single pixels with $2\,e^-$. Dark charge and rounding miscounts produce about 0.5 of them per 1.7-Mpix image at $5 \times 10^{-4}\,e^-$/pix ($\lambda^2/2$ plus $\lambda\,p$), indistinguishable from a real 2-e^- event; two neighbouring single electrons (about 14 per image at $10^{-3}\,e^-$/pix) likewise. Rare-event searches therefore do not decide per event: they keep every candidate and compare the counts with the expected background, which is what the candidate catalogue provides.

19 The few-electron candidate catalogue (26 October)

A separate stage after A_GMX (`clustering_tests/fe_candidates.py`), in the style of the low-background experiments (DAMIC-M, SENSEI). For electron-counting im-

ages (read noise $< 0.35\,e^-$) `run_AGMX` and `run_WADERs --clusterer agmx` write `fe_candidates/<image>_ext<n>_fe_candidates.csv` and `.json`:

- **candidates**: every group of touching pixels with 2–20 counted electrons (≤ 9 pixels): position, charge, pixels, local dark charge and read noise, the statistic T, n_{exp} = the expected number of dark-charge groups per image at least as unlikely in the clean area (simulated blanks; analytic tail $N_{\text{clean}}e^{-T}$ beyond them) and $p_{\text{image}} = 1 - e^{-n_{\text{exp}}}$;
- **mask flags** (nothing is removed): halo within $5\sqrt{E/\text{keV}}$ px (5–60 px) of an A_GMX cluster, charge-transfer trails along its row and column ($E > 1$ keV, 50 px), hot columns (also from an excess single-electron rate), the edge (5 px), overlap with an A_GMX cluster; `clean` = no flag;
- **summary**: the isolated single-electron rate, the clean area, and per class (charge, pixels) the observed clean candidates and the number expected from dark charge alone (blanks simulated from the image's own detection background).

dark charge [e^-/pix]	$2\,e^-$ in 1 px	$2\,e^-$ in 2 px	$3\,e^-$ in 3 px
0.0005	5 / 5.2	58 / 48.2	0 / 0.1
0.001	14 / 12.7	120 / 114.8	0 / 0.7
0.003	89 / 93.4	731 / 734.0	11 / 13.4

Calibration: observed / expected clean candidates in 10 blank images (1.7 Mpix each) per dark charge. Injected 3-e^- events are listed in 99 % of cases (median $n_{\text{exp}} = 0.008$ per image); candidates next to a 60-keV deposit carry the halo flag.

20 Exact pixel integrals (phase 2, 26 October)

The approximation it replaces. Step 2 evaluates each cloud at the pixel centre with its covariance increased by $\frac{1}{12}$ (the variance of a uniform distribution on the pixel), and the M-step subtracts $\frac{1}{12}$ again. This is exact for the second moment but not for the profile: for a cloud with $\sigma \lesssim 0.3$ px almost all the charge is in one or two pixels, the approximated peak is too high and the tails too low, and the EM compensates by moving charge between neighbouring clouds. The visible symptom was a sharp (front-side) X-ray next to a diffuse one: the sharp one was lost or merged in 22 % of cases and its energy was within 5 % only 35 % of the time.

The exact model. A cloud with covariance Σ_k deposits in pixel $p = (i, j)$ (pixel centres at integers) the charge

$$g_k(p) = A_k \int_{i-\frac{1}{2}}^{i+\frac{1}{2}} \int_{j-\frac{1}{2}}^{j+\frac{1}{2}} \mathcal{N}\big(\mathbf{x}; \boldsymbol{\mu}_k, \Sigma_k\big)\, d\mathbf{x} \;=\; A_k \Big[\Phi\big(\tfrac{i+\frac{1}{2}-\mu_y}{\sigma_y}\big) - \Phi\big(\tfrac{i-\frac{1}{2}-\mu_y}{\sigma_y}\big)\Big] \Big[\Phi\big(\tfrac{j+\frac{1}{2}-\mu_x}{\sigma_x}\big) - \Phi\big(\tfrac{j-\frac{1}{2}-\mu_x}{\sigma_x}\big)\Big], \tag{23}$$

the product of two one-dimensional pixel integrals (error functions). The product form is exact only for an uncorrelated cloud; it is used for clouds with $|\rho| = |\sigma_{xy}|/\sqrt{\sigma_{yy}\sigma_{xx}} < 0.5$ (all X-rays), and the strongly correlated clouds (pieces of tilted tracks, which are broad along the track) keep the Gaussian with $+\frac{1}{12}$. The footprint is unchanged (4σ ellipse, at least 2 px).

A consistent M-step. The weighted moments of a cloud's share are now the mean and variance of the *pixelised* Gaussian, which depend on σ and on the sub-pixel position of the centre (a $\sigma = 0.2$ px cloud centred on a pixel has variance $0.01\,\text{px}^2$, centred on a pixel edge $0.25\,\text{px}^2$). The

M-step inverts them with tables computed once (variance against σ^2 and the centre offset; mean against offset and σ^2): $\sigma_y^2 = V^{-1}\big(\mathrm{var}_y \,|\, \mathrm{frac}(\mu_y)\big)$, $\mu_y = \bar{M}^{-1}\big(\bar{y} \,|\, \sigma_y^2\big)$, the same for x. Without this step the exact model and the moment update disagree, and sharp clouds drift in width.

Cost. The compiled EM pass (section 16) has an exact kernel (`math.erf`); it agrees with the numpy version to 10^{-15} keV. The time per image increases by $\approx 20\,\%$ (the prototype without the compiled kernel took twice as long).

Two settings that change with it. (i) The island arbitration (section 6.5, tracks against several X-rays) compares the BIC of fixed-shape clouds with free clouds; the exact model fits sharp X-rays with free clouds better, so the margin was recalibrated on the simulation below from 30 to 100. (ii) With the better fit a sharp X-ray is one cloud instead of three, and a faint fragment (a few electrons in the tail of a large deposit, fitted as a thin "track" cloud) could then join both the X-ray and the deposit and merge them. A fragment — a cloud below $50\,e^-$ and below 10 % of its brightest merge partner — now joins only that partner's cluster: it can be part of a cluster but cannot connect two. (The first version used only the 10 % condition; next to a 600-keV deposit a 5000-e^- piece of the same deposit then counted as faint and the deposit was split, +25 % clusters on the crowded CCD.)

The reported width. The width that separates front- and back-side events is the second moment of the cluster's shared charge minus $\frac{1}{12}$ (audit D2). It is unbiased for $\sigma \geq 0.4\,\mathrm{px}$, but below it the pixelised variance is so small that the noise and dark charge in the footprint dominate it; inverting it exactly does not help (0.28 for a 0.2-px X-ray). The exact-pixel EM width is unbiased there, so a single cloud with an EM width below 0.4 px reports it. Single 5.9-keV X-rays (80 per width, $0.15\,e^-$ noise):

true σ [px]	0.2	0.3	0.4	0.6	1.0	1.5
moment $-\frac{1}{12}$ (before)	0.168	0.289	0.400	0.600	0.997	1.505
reported now	0.250	0.304	0.401	0.600	0.997	1.505
$E_{\mathrm{rec}}/E_{\mathrm{true}}$ (both)	1.000	1.000	1.000	1.000	1.000	1.000

A 0.2-px cloud puts 99 % of its charge in one pixel when centred, so its width is barely measurable. Either way it is far below the front/back cut at 1 px.

Simulation: sharp X-rays next to diffuse ones. Skipper blanks ($0.1\,e^-$/pix, $0.16\,e^-$ noise), 5.9-keV X-rays in four scenes, 777 of each (three images):

scene	cluster	found		$\lvert E/E_{\mathrm{true}} - 1\rvert < 5\,\%$	
		before	exact	before	exact
isolated sharp (σ 0.15–0.35)	sharp	1.000	1.000	1.000	1.000
sharp 3 px from diffuse (σ 1.2–1.6)	sharp	0.781	0.988	0.350	0.967
	diffuse	0.773	0.981	0.349	0.967
two sharp 3 px apart	sharp	0.942	0.934	0.829	0.851
isolated diffuse (σ 1.0–1.6)	diffuse	1.000	1.000	1.000	1.000

The energies of the separated pairs are within ±0.8 % (median). Isolated clusters do not change.

Real data. The wide-test subset (every 12th Fe-55 image-CCD, every 6th Am-241; 80 paired image-CCDs, the earlier injection placement), current model against exact pixel integrals:

	before	exact
Fe-55: injected X-ray pairs 4 px apart resolved	0.907	0.941
Fe-55: injected X-rays / muons / worms correct	0.956 / 0.640 / 0.839	0.958 / 0.640 / 0.839
Am-241: injected pairs resolved	0.607	0.655
real front-side Kα (exact − before, median over image-CCDs)	−0.2 eV (68 %: −1.8 to +2.8)	
Kα clusters: χ^2 flag median (90th percentile)	6.46 (33.3)	1.31 (4.18)
Kα clusters: front-side ($\sigma_{xy} < 1$) fraction	0.591	0.587
time per image-CCD, Fe-55 / Am-241 (median)	0.8 / 42 s	0.9 / 62 s

The model now describes the X-rays it fits (the χ^2 flag falls five-fold), so that flag separates contaminated clusters better; its cut has to be recalibrated. The real front-side Kα does not move.

The crowded CCD. The 29 Am-241 images of the crowded CCD (AMtest EXT4) with the constrained line fit of section 14 (A_GMX, $\sigma_{xy} < 0.7$, blend < 0.02):

	clusters (all)	k	dev/ndf	Z (Np L, γ 26.3)
published (section 14)	—	1.0014	1.98	34.3
before (with the hot-column fix below)	497 765	1.0016 ± 0.0005	1.75	32.8
exact pixel integrals	504 440	1.0006 ± 0.0003	2.59	37.4
+ ped < 10		0.9990 ± 0.0003	2.12	29.9 (before 26.8)

The lines are more significant (Lβ 19.9 → 22.4 σ, Lγ 17.9 → 20.6, γ 26.3 18.3 → 21.1) and the energy scale is closer to 1, but the deviance per degree of freedom of the spectrum is higher (1.75 → 2.59): the shape between the lines is described less well. This is not understood yet and is an open item.

A hot-column bug found on the way. On this CCD the hot-column correction of the audit (B1) flagged 2400 of the 3348 columns as hot, with $\approx 17\,e^-$/pix: its clip was iterated with a threshold that grew with the excess it measured, so on a charge sheet it swallowed the event charge. Every energy was pulled down by ≈ 0.5 % (59.5-keV peak 59.55 → 59.25 keV on one image; $k = 0.975$ on the 29 images). A hot column is now a single column standing out from its ±10 neighbours (one clip around the column median), and nothing is flagged when more than 10 % of the columns would be; on this CCD no column is flagged, and the published numbers are recovered (table above, "before").

EM convergence. The EM runs a fixed 30 iterations (section 6.5). Running it to convergence (largest centre step < 0.01 px, at most 300 iterations; 99.4 % of the fits converged) was tested with A_GMX on the 29 crowded images (before the hot-column fix, so k is low in both): $Z = 6.2 \rightarrow 4.3$ with the blend cut and 6.0 → 7.2 with the ped cut in addition, dev/ndf 1.28 → 1.04 and 1.51 → 1.41. There is no consistent gain, and the cluster energies were already stable at 30 iterations (Step 2). **The fixed 30 iterations are kept.**

Switch. `AGMX_PIX_EXACT=0` restores the earlier pixel model, island margin 30 and merge graph.

21 Does A_GMX supersede the other methods?

Yes, as the general clusterer. Against each alternative:

method	where it is still better than A_GMX	verdict
MHC	Fe-55 data sets: Fe-55 success 0.964 vs 0.951 (energies equal); hot columns 0.78 vs 0.74; 4-px pairs on Am-241 sets 0.83 vs 0.62 (A_GMX hands the noisy PD06 images to hysteresis)	**superseded.** A_GMX is better on crowded images (Lβ 19.9 vs 8.0σ), noisy images (0.86 vs 0.61), high dark charge, worms, alphas, 3-px pairs (0.75 vs 0.29), diffuse 5-px pairs (0.93 vs 0.77) and energy resolution (0.19 vs 0.21 %), and is faster (2× on typical images, 3× on the Am-241 frames, ∼300× on the noisy PD06 frames). MHC's model-selection idea lives on in GMX's island test. Keep the code as a cross-check.
hybrid v1 / v2	nothing significant	**superseded** (they merge diffuse pairs and fail on crowded CCDs).
adaptive (MHC engine)	nothing significant	**superseded**: same switch, weaker engine (ties WADERs on the crowded CCD instead of beating it).
GMX alone	identical except on "unmodelled" images, where it fails (Fe-55 0.49)	**superseded** by A_GMX (GMX + the switch).
MFD+G	isolated X-rays only: Fe-55 success 0.972 vs 0.951, 5× faster	**keep for Fe-55 gain calibration** (energies identical; only point-like X-rays matter). Not for tracks (muon 0.18).
B auto	as fast as WADERs (100× faster than A_GMX); better than WADERs everywhere except the noisy PD06 images	**keep** as the cheap pipeline option (`--cluster-threshold-e auto`).
WADERs	noisy "unmodelled" images (where A_GMX *is* WADERs); speed; the established pipeline	**keep as the reference**; it is what A_GMX falls back to.

In one line: use A_GMX for analysis (it is never much worse than the best method in any regime and is the best in most); use MFD+G when you only need the Fe-55 peak fast; keep WADERs / B for the standard pipeline until A_GMX is integrated and checked by eye on real tracks.

22 Limits and open items

- **Very sharp clusters** (done in phase 2, section 20): the pixel-integration approximation ($\Sigma + \frac{1}{12}$) over-predicted the peak for $\sigma \lesssim 0.3$ px, and a sharp X-ray next to a diffuse one was lost or mis-shared; the exact pixel integral fixes it (found $0.78 \to 0.99$, within 5 % $0.35 \to 0.97$) for +20 % time. Widths below ≈ 0.25 px remain hard to measure (a centred 0.2-px X-ray is one pixel).
- **Unmodelled images** fall back to WADERs: close pairs are then not resolved (0.15). A background model for those images would let GMX run there.
- **EM depends on the seeds** (scale-space seeding is what makes close pairs work); no global optimum is guaranteed.
- **Island test** tolerates 5 % non-Gaussianity; only checked on injected objects and Fe-55. **Real Am-241 tracks have not been inspected by eye.**
- **Labels:** muon/worm swap ∼30 %; classes only validated on injections.
- **Crowded fit quality:** dev/ndf 1.93 with structure at 25–28 keV; the σ_{xy} cut is not re-optimised for A_GMX's width.
- **Speed:** median 0.7 s, but minutes on the largest crowded CCDs (WADERs: < 1 s).

- Not integrated in `run_WADERs` (would be an optional flag; needs approval).

23 Parameters, code, how to run

parameter	default	where	meaning
`excess_ratio`	3	adaptive	unmodelled if $s \geq 3\sigma_{\text{read}}$ (and $\geq 0.5\,e^-$)
`occ_thr`, `box`	0.3, 31 px	adaptive	"crowded" label (information)
`scales`	0.5, 1, 1.5, 2.5 px	GMX	matched-filter widths
`nsig`	5	GMX	seed significance (+ Poisson check at the same tail)
`n_iter`	30 (+15 residual, +5 pruning)	GMX	EM iterations
`nsig_res`	5	GMX	residual-pass seeds
`sig_min`	4	GMX	prune clouds with S/N below
`r_merge`	0.5	GMX	ridge ratio for merging
`w_ratio`	1.6	GMX	width ratio (sharp-on-diffuse protection)
`link_alpha`, `island_max`, `bic_margin`	0.01, 4000 px, 30	GMX	island test
`f_sat`, `sat_e_max`	0.01, 1 keV	GMX	satellite absorption (only fragments < 1 keV)
`ped_ring`, `ped_q`	(5, 8) px, 25	GMX	local pedestal flag (ring, percentile)
`grow`	3 px	GMX	CTI-trail growth
`p_count`, `count_E_max`	10^{-5}, $100\,e^-$	GMX	few-electron finder, counting energies

```
import sys; sys.path.insert(0, "comm_scripts/clustering_tests")
import adaptive as AD
clusters = AD.adaptive_cluster(Q_electrons, valid_mask, engine="gmx")
# each cluster: E [keV], y, x, sxy [px], cls, blend, sig, n_comp, edge, ped [e-/pix],
    peakfrac, chi2,
# regime ('sparse' / 'crowded' / 'unmodelled'), occ
# spectroscopy selection (section on contamination):
good = [c for c in clusters if c.get("blend", 0) < 0.02 and not c.get("edge")
       and not c.get("ped", 0) >= 10 # not sitting on a charge sheet (crowded fields)
       and c.get("peakfrac", 1) > 0.95] # not merged with a neighbouring deposit
```

Test scripts (all in `clustering_tests/`, README there): `wide_test.py` + `wide_run.py adaptive+GMX` (all images), `adaptive_validate.py` (tracks, few-electron, Fe-55, crowded), `make_agmx_figures.py`, `make_agmx_perf.py`. Unit tests: `python -m unittest tests.test_clustering_methods tests.test_agmx_root`.

23.1 Running A_GMX on WADERs data: cluster files in the WADERs format

A_GMX is part of the `run_WADERs` chain as an option, and can be run on a finished run:

```
run_WADERs images.txt --gain-correction --clusterer agmx --composite # A_GMX files,
    composite on them
run_WADERs images.txt --gain-correction --clusterer both --composite # also
    composite_agmx/ next to composite/
run_AGMX WADERs_analysis_<run> [-j 4] [--per-image] [--no-gaincorr] # a run that is
    already finished
```

It runs on the calibrated image panaSKImg saves for every CCD (`<image>_EXT<n>_waders.fits.bz2`). With the gain correction, the image is first converted ADU $\rightarrow e^-$ with the same $g(n)$ curves and anchor as `root_gaincorr/`; a CCD without a reliable curve keeps its fitted gain, as there. The output is `root_agmx/` and `root_agmx_gaincorr/`: one `panaSKImg_clustersRec_<image>_ext<n>.root` per CCD, written by `src/agmx_root.py`. Every A_GMX cluster (its pixels and the charge it takes from each, which sums to E) becomes a pysimdamicm `Cluster` object, the ClusterFinder's own class, so all branches (positions, `wSTD`, `fSTD`, PCA, track end points, seed, `closest_cls_*`, `Energy_PS/AVG`, …) are computed by

panaSKImg's code. `info` and `process_config` are copied from the CCD's panaSKImg file, and the extra branches `agmx_cls`, `agmx_blend`, `agmx_ped`, `agmx_peakfrac`, `agmx_chi2`, `agmx_sig`, `agmx_regime` and a `TNamed agmx_config` are added. MakeComposite and every other reader of WADERs files read them unchanged.

Check. `clustering_tests/agmx_root_check.py` writes the clusters of real panaSKImg files (EXT1, EXT3 and EXT4 of an Am-241 image: 1652, 7686 and 5452 clusters) through the writer and compares every branch. They are identical except in numerically degenerate cases, where panaSKImg's own result depends on the last bits of its inputs:

- Gaussian fits (`fSTD`) of profiles 1–3 bins wide;
- the sign of a PCA eigenvector (θ versus $\theta \pm \pi$);
- PCA of 1–2 pixel clusters;
- `closest_cls_*` when the nearest cluster lies in the same row or column, because panaSKImg excludes "itself" with `PosX != ... and PosY != ...`.

With the default `--clusterer waders` the pipeline output is identical to before (`utils/compare_runs.py`). A flag value of -999 in the files means "not computed" (e.g. `agmx_chi2` of a cluster the mixture did not fit). Every cluster carries the flags: clusters from the few-electron finder and the hysteresis branch get `edge` and `ped` like the others, `blend` = 0 and `peakfrac` = 1 (no charge is shared).

Fixes of 25 October (audit). (1) The peak search ignored masked pixels only after the local-maximum test, so the matched-filter significance just outside the mask (variance 0 there) could hide a peak on the first or last valid row or column: such an X-ray came out as a flag-less few-electron cluster. The local maximum is now taken over valid pixels only, and the X-ray is a normal cluster with `edge` set. (2) The Poisson check of residual few-electron seeds removed the dark charge twice; with it removed once, a simulation at $\lambda = 0.5\,e^-$/pix finds 20 instead of 14 of 60 injected $15\,e^-$ events, with no clusters on pure-noise images either way. Both change only events touching the mask and high-dark-current CCDs. All performance numbers of this document were re-run with the fixed code (wide test, Fe-55, tracks, few-electron and crowded-CCD validations; results in `*_v3` folders of the data directory). Against the earlier numbers: wide-test Fe-55 success 0.925 $\rightarrow$ 0.927, 4-px pairs 0.87 $\rightarrow$ 0.88, Am-241 data sets 0.819 $\rightarrow$ 0.824; on the denser CCD EXT3 hot columns 0.28 $\rightarrow$ 0.21 and 6-px pairs 1.00 $\rightarrow$ 0.98; few-electron efficiencies and fakes unchanged; crowded CCD with the blend cut Z 34.3 $\rightarrow$ 34.8 (k 1.0014 $\rightarrow$ 1.0016).

A All performance tables

Generated by `clustering_tests/make_agmx_perf.py` from the saved results (same tests as section 14; all methods; "ok" = one cluster with energy within 5 %, pairs: both resolved).

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.856	0.52 %	0.59	0.74	0.11	0.01
B auto	0.905	0.59 %	0.62	0.78	0.13	0.01
MFD+G	0.933	0.21 %	0.18	0.37	0.94	0.14
hybrid v1	0.873	0.20 %	0.58	0.66	0.84	0.02
hybrid v2	0.930	0.21 %	0.63	0.68	0.85	0.05
MHC	0.933	0.21 %	0.64	0.77	0.86	1.61
adaptive (MHC engine)	0.931	0.21 %	0.64	0.77	0.83	1.26
GMX	0.916	0.19 %	0.63	0.78	0.89	0.69
A_GMX	0.927	0.19 %	0.64	0.79	0.88	0.69

Wide test, all (878 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.880	0.47 %	0.62	0.78	0.11	0.01
B auto	0.927	0.53 %	0.65	0.81	0.13	0.01
MFD+G	0.972	0.17 %	0.20	0.39	0.95	0.13
hybrid v1	0.926	0.16 %	0.63	0.71	0.87	0.02
hybrid v2	0.972	0.17 %	0.68	0.73	0.88	0.04
MHC	0.964	0.17 %	0.68	0.81	0.87	0.94
adaptive (MHC engine)	0.961	0.17 %	0.67	0.81	0.87	0.83
GMX	0.954	0.16 %	0.67	0.83	0.91	0.59
A_GMX	0.954	0.16 %	0.67	0.83	0.91	0.60

Wide test, Fe-55 data sets (705 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.830	0.77 %	0.42	0.60	0.12	0.11
B auto	0.822	0.89 %	0.42	0.60	0.14	0.24
MFD+G	0.738	1.21 %	0.11	0.27	0.88	3.79
hybrid v1	0.605	0.78 %	0.29	0.37	0.69	8.10
hybrid v2	0.728	1.18 %	0.37	0.46	0.72	16.89
MHC	0.793	1.11 %	0.41	0.53	0.83	76.69
adaptive (MHC engine)	0.794	0.95 %	0.41	0.56	0.61	38.31
GMX	0.736	0.75 %	0.38	0.50	0.75	40.84
A_GMX	0.824	0.72 %	0.42	0.58	0.63	23.31

Wide test, Am-241 data sets (115 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.567	1.59 %	0.50	0.53	0.10	0.01
B auto	0.784	1.58 %	0.64	0.76	0.18	0.03
MFD+G	0.824	1.18 %	0.14	0.29	0.97	0.32
hybrid v1	0.738	1.05 %	0.52	0.60	0.79	0.07
hybrid v2	0.804	1.17 %	0.55	0.58	0.81	0.21
MHC	0.817	1.17 %	0.68	0.72	0.86	10.70
adaptive (MHC engine)	0.812	1.17 %	0.68	0.71	0.85	7.77
GMX	0.794	1.17 %	0.69	0.76	0.88	1.30
A_GMX	0.794	1.17 %	0.69	0.76	0.88	1.30

Wide test, skip-noise study (58 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.867	1.66 %	0.48	0.66	0.08	2.40
B auto	0.802	2.08 %	0.39	0.65	0.15	2.46
MFD+G	0.281	9.15 %	0.08	0.09	0.73	69.85
hybrid v1	0.275	8.24 %	0.13	0.10	0.67	98.38
hybrid v2	0.278	8.99 %	0.15	0.10	0.68	156.01
MHC	0.608	5.62 %	0.45	0.45	0.81	396.59
adaptive (MHC engine)	0.864	2.02 %	0.50	0.70	0.15	1.29
GMX	0.491	4.42 %	0.31	0.36	0.65	193.75
A_GMX	0.864	2.02 %	0.50	0.70	0.15	2.38

Wide test, noisy (bg spread $\geq$ 1 e-) (27 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.550	2.40 %	0.33	0.45	0.07	0.08
B auto	0.806	2.10 %	0.48	0.66	0.17	0.08
MFD+G	0.772	1.07 %	0.15	0.29	0.91	0.70
hybrid v1	0.724	1.02 %	0.39	0.51	0.82	0.11
hybrid v2	0.765	1.06 %	0.44	0.52	0.83	0.32
MHC	0.829	1.00 %	0.54	0.64	0.88	16.18
adaptive (MHC engine)	0.868	0.94 %	0.55	0.68	0.75	7.31
GMX	0.803	0.88 %	0.52	0.67	0.88	2.05
A_GMX	0.869	0.90 %	0.56	0.74	0.79	1.97

Wide test, dark $\geq$ 1 e-/pix (152 image-CCDs)

method	Fe-55 ok	Fe-55 E rms	muon	worm	pair 4 px	s / image-CCD (median)
WADERs	0.590	1.64 %	0.27	0.43	0.08	0.08
B auto	0.781	1.60 %	0.39	0.58	0.15	0.18
MFD+G	0.789	1.05 %	0.12	0.29	0.90	1.55
hybrid v1	0.684	0.84 %	0.32	0.44	0.74	0.36
hybrid v2	0.781	1.02 %	0.39	0.50	0.77	2.43
MHC	0.822	0.98 %	0.44	0.58	0.85	25.13
adaptive (MHC engine)	0.819	0.92 %	0.44	0.59	0.70	11.35
GMX	0.791	0.79 %	0.42	0.60	0.83	9.08
A_GMX	0.845	0.77 %	0.45	0.65	0.76	2.50

Wide test, occupancy ≥ 30 % (194 image-CCDs)

method	sigma 0.3	0.8	1.4	1.8	pair 5 px resolved	real Ka / 5.899	Ka spread
WADERs	0.9974 (0.990)	0.9843 (0.943)	0.9521 (0.837)	0.9214 (0.737)	0.00	0.9928	1.32 %
MFD+G	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9999 (0.995)	0.78	0.9968	0.29 %
MHC	1.0001 (1.000)	1.0002 (0.999)	1.0001 (0.996)	0.9998 (0.995)	0.77	0.9969	0.26 %
GMX	1.0001 (1.001)	0.9999 (0.998)	0.9996 (0.994)	0.9997 (0.994)	0.93	0.9969	0.23 %
A_GMX	1.0001 (1.001)	0.9999 (0.998)	0.9996 (0.994)	0.9997 (0.994)	0.93	0.9969	0.23 %

Fe-55 injection (DM09 / DM10, EXT1 / EXT4, images 0-1): median E_rec/E_true (worst CCD) per width, pairs, real Ka

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.90	0.93	0.94	0.97	0.97	0.90	0.76	0.47	0.00	0.21	0.71
MFD+G	0.19	0.28	0.97	1.00	1.00	0.78	0.00	0.47	0.29	0.99	1.00
MHC	0.94	0.93	0.97	1.00	1.00	0.78	0.78	0.53	0.29	0.92	1.00
GMX	0.93	0.93	0.96	0.99	0.97	0.94	0.74	0.64	0.75	0.93	0.93
A_GMX	0.93	0.93	0.96	0.99	0.97	0.94	0.74	0.64	0.75	0.93	0.93

Injected objects in real Am-241 images, EXT1 (3 images x 3 seeds, success fraction)

method	muon	worm	blob	single	fe55	alpha	hotcol	lowE	pair3	pair4	pair6
WADERs	0.17	0.81	0.84	0.98	0.97	0.28	0.32	0.17	0.00	0.21	0.90
MFD+G	0.02	0.24	0.92	1.00	0.98	0.10	0.02	0.34	0.22	0.98	1.00
MHC	0.20	0.79	0.92	1.00	0.98	0.10	0.28	0.28	0.22	0.92	1.00
GMX	0.20	0.81	0.89	0.97	0.97	0.21	0.25	0.40	0.78	0.98	0.98
A_GMX	0.20	0.81	0.89	0.97	0.97	0.21	0.25	0.40	0.78	0.98	0.98

Injected objects in real Am-241 images, EXT3 (3 images x 3 seeds, success fraction)

dark e-/pix	method	2 e-	3 e-	4 e-	5 e-	7 e-	fakes/Mpix
0.001	WADERs	0.00	0.00	0.00	0.11	0.29	0.0
0.001	MHC	0.01	0.81	0.97	0.99	1.00	0.0
0.001	GMX	0.01	0.81	0.98	1.00	1.00	0.0
0.001	A_GMX	0.01	0.81	0.98	1.00	1.00	0.0
0.001	DM-style ≥1e-	1.00	1.00	1.00	1.00	1.00	7.3
0.01	WADERs	0.00	0.00	0.00	0.11	0.30	0.0
0.01	MHC	0.00	0.01	0.38	0.93	1.00	0.0
0.01	GMX	0.00	0.01	0.38	0.93	1.00	0.0
0.01	A_GMX	0.00	0.01	0.38	0.93	1.00	0.0
0.01	DM-style ≥1e-	1.00	1.00	1.00	1.00	1.00	462.3
0.05	WADERs	0.00	0.00	0.00	0.11	0.35	0.0
0.05	MHC	0.00	0.00	0.00	0.04	0.56	0.0
0.05	GMX	0.00	0.00	0.00	0.04	0.56	0.0
0.05	A_GMX	0.00	0.00	0.00	0.04	0.56	0.0
0.05	DM-style ≥1e-	1.00	1.00	1.00	1.00	1.00	7794.0

Few-electron events on simulated blanks (efficiency by event charge; fakes = clusters of 2-5 e- per Mpix with nothing injected)

clusters	N	k	dev/ndf	Np La	Np Lb	Np Lg	gamma 26.3	Ag K	Z (Np L + 26.3)
WADERs (sigma_xy < 0.7)	62665	0.9840 ± 0.0008	1.74	4.0	15.5	16.0	14.8	12.9	27.1
A_GMX (sigma_xy < 0.7, blend < 0.02)	82073	1.0016 ± 0.0004	1.93	3.7	20.8	19.2	19.9	18.1	34.8

Crowded Am-241 CCD (runs_29img EXT4, 29 images): constrained 10-30 keV fit, line significances [sigma]

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